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Assessing the Impact of U.S. Air Force National Security Space Launch Acquisition Decisions

2023 Update



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About This Report

The U.S. National Security Space Launch program is responsible for launching the military and intelligence community satellites that serve as the global eyes, ears, and timekeepers of the nation. In 2019, the U.S. Air Force tasked RAND with providing analysis of the global heavy lift launch market to inform the Air Force's National Security Space Launch Phase 2 acquisition. Decisions of foreign launch providers in the international market and the emergence of new domestic launch service providers in the years after 2019 may provide opportunities for the U.S. Space Force (USSF) to better support the U.S. launch industrial base and provide warfighters with increased flexibility in the contested space domain. Therefore, as the USSF embarks on Phase 3 acquisition for the NSSL, it asked RAND to explore how changes to the market affect the findings and recommendations of the prior analysis. The intended audience for this report includes USSF leadership, the Department of the Air Force, and the U.S. Congressional Armed Services and Appropriations Committees.

The research reported here was commissioned by the USSF's Program Executive Office for Assured Access to Space and conducted within the Force Modernization and Employment Program of RAND Project AIR FORCE as part of a fiscal year 2023 project, "Space Launch Market Research and Assessment."

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This report documents work originally shared with the DAF in a peer-reviewed, annotated presentation on August 14, 2023. In early 2024, the analysis was updated to include calendar-year 2023 launches; and a draft report, dated February 2024, was provided to the DAF. This final report was reviewed by formal peer reviewers and DAF subject-matter experts.

The view expressed in this report are those of the author and do not reflect the official policy or position of the Department of Defense or the U.S. Government.

Review of this material does not imply Department of Defense endorsement of factual accuracy or opinion.

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Summary

Issue

The U.S. military depends on space-based assets for strategic and tactical communications, for collection of imagery over contested areas, for warning of missile launches, and for precision navigation and timing. The ability to replenish those assets in a timely fashion may affect the overall readiness of the force. For this reason, the U.S. Space Force (USSF) is required to sustain at least two viable U.S.-based launch service providers for national security space (NSS) payloads. In 2019, a RAND team assessed the global heavy lift launch market to understand the impact of NSS launch acquisition decisions on that market. Space Exploration Technologies Corporation (SpaceX) and United Launch Alliance (ULA) are the current NSS launch providers, having been selected for the Phase 2 launch contracts awarded between fiscal years 2020 and 2024. Although ULA is retiring its prior NSS-certified launch vehicles (Atlas V and Delta IV Heavy), its new vehicle (Vulcan) has yet to complete NSS certification, leaving the USSF temporarily reliant on SpaceX as the only NSS-certified launch service provider. Given events over the past four years, the RAND team was asked to update our assessment of the launch markets to better understand how the USSF might shape future acquisition strategies or prepare for operational risks introduced by past acquisition decisions.

Approach

To conduct the assessment, we examine historic and projected levels of supply and demand in the global and NSS heavy lift launch markets. Using these data, we forecast demand over the next ten years (2024–2033) and use Monte Carlo simulations to explore how assumptions regarding future market dynamics might affect the USSF's ability to meet NSS demand and sustain at least two certified suppliers, ideally with support from the commercial markets.

Key Findings

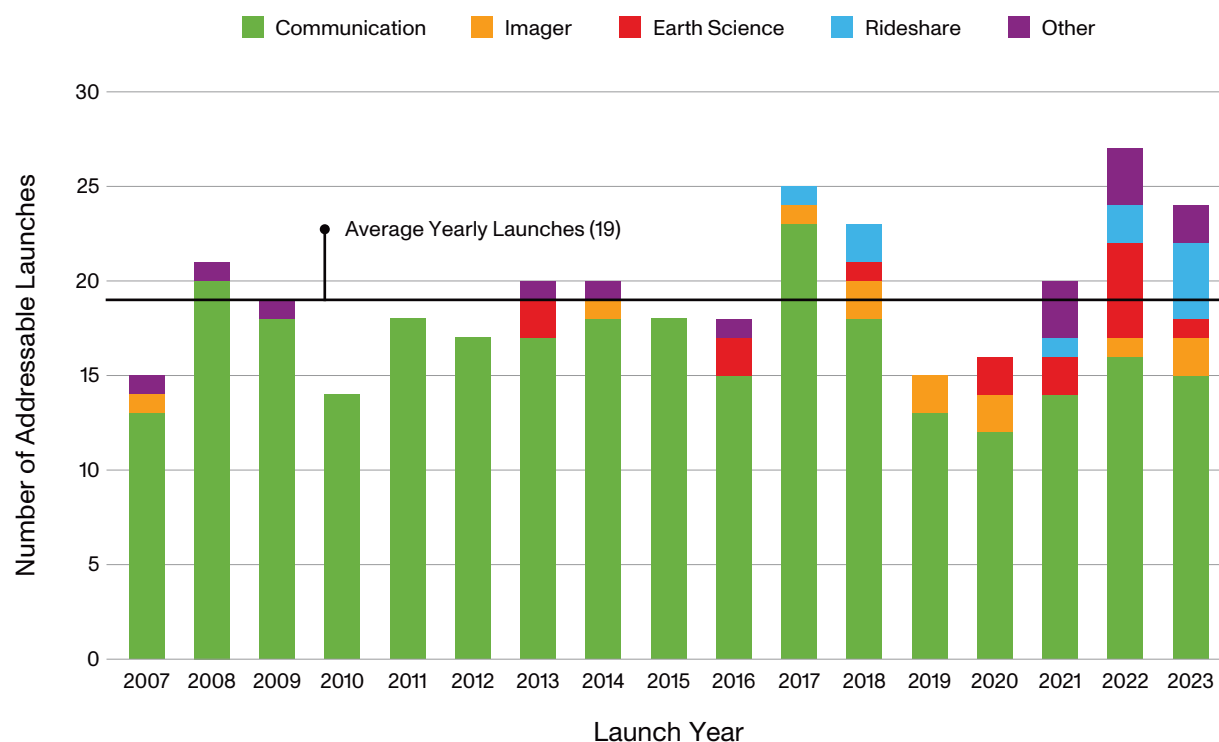
- The average yearly global launch demand addressable by U.S. firms is largely unchanged since the 2019 assessment; while demand is diversifying (see Figure S.1), we expect to see only slight overall growth in the next five years.
- There are few viable competitors in the global addressable launch market; SpaceX dominates with a 70-percent share of the addressable demand in 2022 (compared with a 40-percent share in 2019).
- Given the current lack of competitors, several firms are seeking to enter the heavy lift launch market; we expect rapid consolidation in the market should any be successful.
- Despite the diversity offered by the possible new entrants, there is little diversity in the key resources (fuel, launch complexes, and engine suppliers) needed for launch.

- NSS Phase 2 awards to ULA’s Vulcan launch vehicle and subsequent delays in Vulcan’s development have created a significant backlog; the last payloads awarded under the Phase 2 contract are likely to experience a two-and-one-half-year delay as this backlog is serviced.¹

Recommendations

- The USSF should make prudent preparations for a future that has only two U.S. providers of NSS-certified heavy lift launch, at least one of which may have little support from the addressable marketplace.²
- The USSF should continue to provide support to enable three U.S. providers to enter and/or continue in the heavy lift launch market.³
- The USSF should recognize and mitigate the risks of adding to backlogs of firms that have not yet been certified for NSS launch.
- The USSF may want to consider adding diversity of launch sites or of fuel sources as a secondary consideration in future launch acquisition source selections.

Figure S.1. The Global Addressable Demand Is Diversifying



SOURCE: RAND analysis of primary payload type, RAND’s Heavy Lift Launch Data Set, 2007–2023.

¹ While multiyear NSS launch delays routinely occur due to payload slips, delays due to launch vehicle supply have occurred less frequently—the last being after the loss of the space shuttle *Challenger* in 1986.

² This was also recommended in the 2019 assessment.

³ This was also recommended in the 2019 assessment.

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Introduction

The U.S. military depends on space-based assets for strategic and tactical communications, for collection of imagery over contested areas, for warning of missile launches, and for precision navigation and timing. The ability to replenish those assets in a timely fashion may affect the overall readiness of the joint force. For this reason, the U.S. Space Force (USSF) is required by statute to sustain at least two viable U.S.-based launch service providers for national security space (NSS) payloads.¹ In 2019, a RAND team assessed the global heavy lift launch market to understand the impact of U.S. Air Force acquisition decisions on that market.² In the years since our original research, Space Exploration Technologies Corporation (SpaceX) and United Launch Alliance (ULA) were selected as providers for the NSS Phase 2 launch contracts awarded between fiscal year (FY) 2020 and FY 2024. Although ULA is retiring its prior NSS-certified launch vehicles (Atlas V and Delta IV Heavy), its new vehicle (Vulcan) has yet to complete NSS certification, leaving the USSF temporarily reliant on SpaceX as the only certified NSS launch service provider.³ Given these events and others over the past four years, the RAND team was asked to update our assessment of the launch markets to better understand how

¹ 10 U.S. Code §2273 states:

- (a) POLICY.—It is the policy of the United States for the President to undertake actions appropriate to ensure, to the maximum extent practicable, that the United States has the capabilities necessary to launch and insert United States national security payloads into space whenever such payloads are needed in space.
- (b) INCLUDED ACTIONS.—The appropriate actions referred to in subsection (a) shall include, at a minimum, providing resources and policy guidance to sustain — (1) the availability of at least two space launch vehicles (or families of space launch vehicles) capable of delivering into space any payload designated by the Secretary of Defense or the Director of National Intelligence as a national security payload; (2) a robust space launch infrastructure and industrial base; and (3) the availability of rapid, responsive, and reliable space launches for national security space programs to— (A) improve the responsiveness and flexibility of a national security space system; (B) lower the costs of launching a national security space system; and (C) maintain risks of mission success at acceptable levels.

The quotation is taken from the 2023 version of the statute; the basic requirements have been in force since the early 2000s. The parenthetical, “(or families of space launch vehicles),” was added in 2008 to accommodate the merger that produced ULA. The USSF has the accountability to fulfill these requirements on behalf of DoD and the intelligence community, as did the U.S. Air Force (USAF) before the creation of the USSF.

² Bonnie L. Triezenberg, Colby P. Steiner, Grant Johnson, Jonathan Cham, Éder M. Sousa, Moon Kim, and Mary Kate Adgie, *Assessing the Impact of U.S. Air Force National Security Space Launch Acquisition Decisions: An Independent Analysis of the Global Heavy Lift Launch Market*, RAND, RR-4251-AF, 2020.

³ This is the status as of August 2024. The 2019 assessment warned that Vulcan was unlikely to meet its predicted first launch date of May 2021. Vulcan’s actual first launch date was January 2024, toward the end of the uncertainty interval used in our analysis. To be certified for NSS launch, the Vulcan chose a certification path that requires it to have successfully completed two launches. The payload originally planned for Vulcan’s second launch has been delayed, and the date for a second launch has not been announced. For background on the NSS certification paths, see Triezenberg et al., 2020.

the USSF might (1) shape future acquisition strategies or (2) prepare for operational risks introduced by past acquisition decisions.⁴

The conclusions of the 2019 study were as follows for the global commercial and NSS launch markets:

- Worldwide, the annual number of launches grew to 71 in 2018 from 47 in 1998, but the commercial portion over which launch firms compete—*addressable share*—stayed steady at around 20 launches (as it does today).
- SpaceX’s Falcon 9 handled more than half of addressable launches in 2019, with communications companies being the dominant customers.
- This market is likely to see only moderate growth over the next ten years, and the U.S. share of it is likely to fall. The net result is that the addressable share of the launch market is unlikely to support more than one U.S. supplier of launch services focused on commercial heavy lift.⁵

The addressable market, in fact, has not grown in the last four years. While, for reasons we will discuss later in this report, the U.S. share of the addressable market has not (yet) fallen, this is a temporary condition and the underlying geopolitical factors that would cause the U.S. market share to fall have not changed. We continue to assess that the long-term addressable share of the launch market is unlikely to support more than one U.S. supplier of heavy-lift launch services.

The conclusions regarding the NSS launch market in 2019 were:

- In many conceivable futures, the USAF’s current acquisition plan is unlikely to provide sufficient supply of NSS launch vehicles certified to carry U.S. NSS payloads.
- The USAF can lower the risk of insufficient supply of NSS launch vehicles by gaining nonpriority access to legacy launch systems and/or by supporting three launch services providers through the 2022–2025 time frame.⁶

In fact, as we will discuss in this report, such a future has come to fruition.

The recommendations from the 2019 study were that, in the near term, the USSF should “continue to provide tailored support through 2023 to enable three U.S. launch service providers to enter and/or continue in the heavy lift launch market,” and in the longer term, it should “make prudent preparations for a future with only two U.S. providers of NSS-certified heavy lift launch, at

⁴ The term *assessment* is used in this report to indicate our interpretation of past events and our projection of future events. Given the deep uncertainties associated with interpreting how events affect the decisions of private firms and forecasting the future actions and decisions of those firms, our assessments are based on observations and judgments and should not be treated as findings, nor do we use them as the basis for recommendations. These judgments are a product of the research team’s deep expertise in space launch, having collectively over 80 years of experience as buyers, sellers, and observers of the space launch market ecosystem. The term *finding* is reserved for the few cases where our evidence and analytic methods are sufficiently strong that we can make definitive statements regarding what has happened in the past, as well as the likelihood of future events.

⁵ Triezenberg et al., 2020, pp. 67–68.

⁶ Triezenberg et al., 2020, p. 68; “nonpriority access” includes launch services obtained by exercising and negotiating contingency or optional clauses within NSS launch contracts, which, in fact, the USSF did. Unfortunately, as we will detail in Chapter 4, these options were insufficient to fully mitigate the risk.

least one of which may have little support from the commercial marketplace.”⁷ Events over the last four years have only reinforced the urgency of these recommendations.

While the 2019 study spent a significant effort to understand the ramifications of a Phase 2 acquisition decision to award launch contracts to two versus three providers, the USSF decided early in the 2023 study that the Phase 3 acquisition strategy would allow for up to three providers for high-assurance missions and unrestricted competition for more risk-tolerant missions.⁸ Therefore, this 2023 update focuses on (1) better understanding trends in the heavy lift launch market and how those trends might affect the USSF’s ability to meet the goals for assured access to space as spelled out in U.S. statute, (2) assessing the impact of the Phase 2 awards on operational risk, and (3) providing a quick look into the resources needed to sustain resilience in NSS launch services.

Note that this report does not attempt to revisit topics, covered in our prior report, regarding the origins and history of the commercial launch market, the factors that drive buyers and sellers in that market, the historic evidence regarding strategies sellers use to position themselves within the market, or the number of U.S. providers the market may be able to support. This report does, however, discuss relevant changes in the market that have occurred in the past four years and any changes we made to the assumptions of our underlying analysis because of those changes. For these reasons, readers who are unfamiliar with either the commercial or the NSS launch markets are encouraged to read our prior report before venturing further.

Definitions Used in This Report

Heavy Lift Launch. The term *heavy lift launch* in this report is defined by the class of launch vehicles capable of taking the heaviest payloads to orbit. As noted in the 2019 assessment, this is a relative, not absolute, definition, since the heaviest payloads have gotten heavier over time, meaning that the heavy lift launch vehicles of 20 years ago are medium lift launch vehicles today.⁹ More specifically, we define a launch vehicle to be capable of heavy lift if it satisfies one of more of the orbital regimes defined in Table 1.1.¹⁰

⁷ Triezenberg et al., 2020, p. xiv. While the USSF did provide some limited support to a third provider in the intervening years, only two providers were selected for the Phase 2 acquisition. Chapter 4 of this report provides our assessment of the impact of those decisions on the U.S. suppliers in the market and on the supply of NSS launch vehicles.

⁸ The more risk-tolerant missions are termed Lane 1 missions in the NSS Launch Phase 3 acquisition strategy, and the higher-assurance missions are termed Lane 2; U.S. Department of the Air Force (DAF), National Security Space Launch Phase 3 Lane 1 Request for Proposal (RFP), Solicitation FA8811-24-R-0001, October 4, 2023a; DAF, National Security Space Launch (NSSL) Phase 3 Lane 2 Request for Proposal (RFP), Solicitation# FA8811-24-R-0002, October 4, 2023, amended November 7, 2023b.

⁹ As noted in the 2019 assessment, small payloads have also gotten smaller as heavy payloads have gotten heavier. This trend continues in 2023, though we note that recent increases in the mass of the Starlink satellites, soon to be the most prolific objects in space, will undoubtedly reverse this trend.

¹⁰ Table 1.1 uses the USSF’s referenced orbits, as defined by the National Security Space Launch program, as a basis for defining the term *heavy lift launch*, and this is consistent with the definition used in the 2019 research. The Federal Aviation Administration, the U.S. launch regulator, does not define launches by class.

Table 1.1. Mass-to-Orbit Definition for Heavy Lift Launch

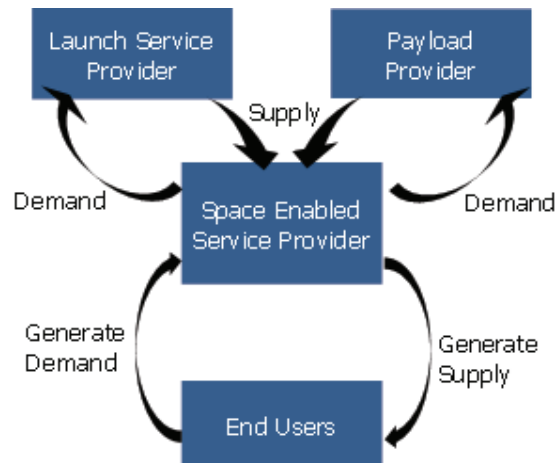
Reference Orbit	Apogee (nmi)	Perigee (nmi)	Inclination (degrees)	Mass to Orbit (lbm)
Low Earth orbit (LEO)	500	500	63.4	15,000
Polar	450	450	98.2	15,500–37,500
Medium Earth orbit (MEO) direct inject 1	9,815	9,815	50	11,750
MEO direct inject 2	10,988	10,988	55	9,000
MEO transfer orbit	10,998	540	55	9,000–11,200
Geosynchronous transfer orbit	19,232	100	27	18,000
Geosynchronous direct inject	19,323	19,323	0	5,000–14,500
Molniya orbit	21,150	650	63.4	11,500

NOTE: lbm = pound-mass; nmi = nautical miles.

Participants in the Launch Ecosystem. Figure 1.1 illustrates the relationships between parties in the overall launch ecosystem. In this view, end users generate demand for space-enabled services (communications, imagery, etc.). The space service provider then generates demand for payloads (i.e., satellites or other objects to be placed into orbit) and for the launch services that are the subject of this report. When the term *provider* is used in this report without a modifier, the reference is to the launch service provider.

As the figure demonstrates, without a demand for space-enabled services, there is no demand for launch services. Only a change in (1) the reasons to go to space or (2) the possibility of turning a profit

Figure 1.1. The Launch Market Ecosystem



SOURCE: Reproduced from Triezenberg et al., 2020, Figure 1.1, p. 3.

based on services supplied from space changes the demand to go to space.¹¹ Therefore, in this report we explore trends to discover possible new sources of demand for space services and the underlying economics of emerging space services when discussing possible growth in the launch markets.

Launch Markets

As in 2019, we recognize two markets in our analysis. The first is the portion of the global heavy lift launch demand that is addressable by U.S. firms, and the second is the NSS launch market. In the global addressable market, both supply and demand can be heavily influenced by international events and by national regulations. A prime example regarding dynamics of supply occurred at the outbreak of war between Russia and Ukraine in February 2022. Prior to the outbreak of the war, the Soyuz launch vehicle (of Russian origin) had a sizable share of the global addressable market. That share vanished virtually overnight after the head of Roscosmos, the Soyuz supplier, refused to launch a payload consisting of 36 OneWeb satellites unless the government of the United Kingdom divested itself of OneWeb (the UK government had invested \$500M in OneWeb to bring the firm back from bankruptcy).¹² The UK government did not divest, and OneWeb terminated its remaining contracts with Soyuz. To complete the build-out of its constellation of satellites, OneWeb awarded space launch contracts to the Indian Space Research Organisation for its Geosynchronous Satellite Launch Vehicle (GSLV) and to SpaceX for its Falcon 9. On the demand side of the equation, much of the total global market is fenced off by various nations in support of their own national goals.

The NSS launch market is much less dynamic. Demand is often set years in advance and, while it can fluctuate based on U.S. national security priorities, it is quite stable. Supply has historically been limited to those launch providers who undergo a lengthy process to become “certified” NSS providers. This will change under the Phase 3 acquisition, which has partitioned the overall NSS demand into two lanes. The first is for more risk-tolerant payloads and will award indefinite delivery/indefinite quantity contracts with no restrictions on the number of providers and minimal qualification requirements.¹³ The second lane is modeled after the Phase 2 procurement to assure access to NSS-certified launch services for NSS high mission assurance payloads.

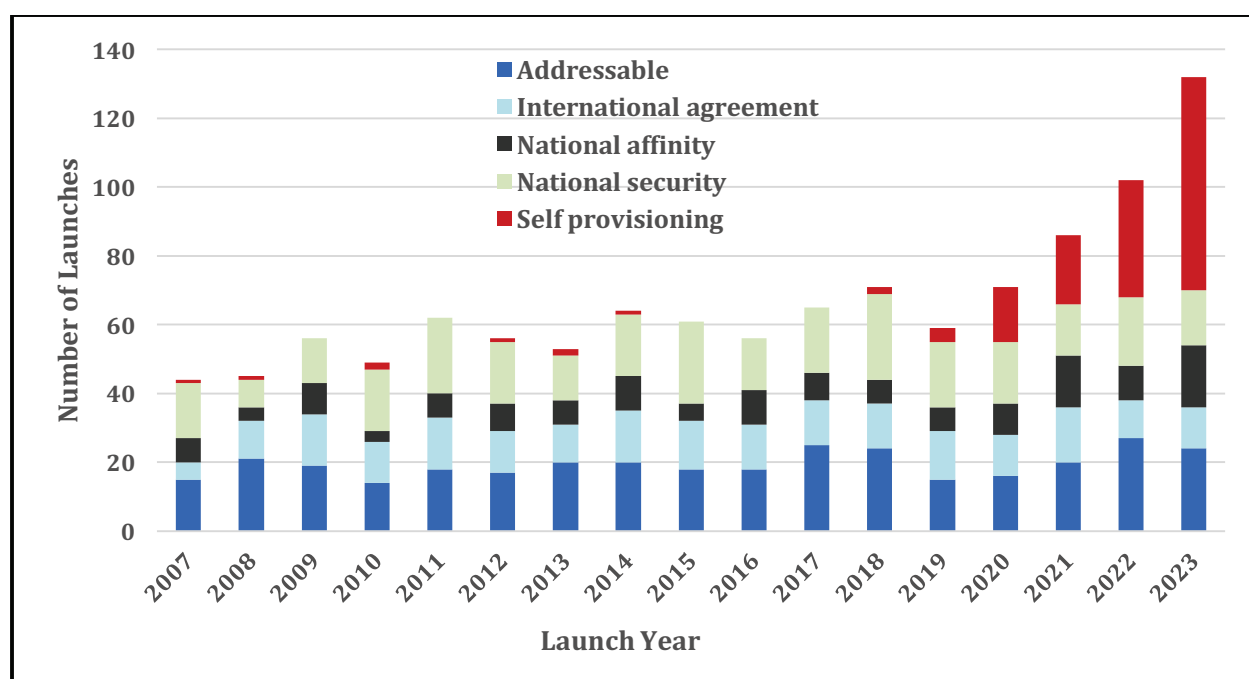
¹¹ A frequent misconception is that lowering the price of launches will make additional space services economically viable, generating additional demand for launch services. For example, in 2000 the keynote address at the International Space University’s summer symposium on space launch stated that a drop in price per launch from \$10,000/kg to \$1,000/kg would result in a tenfold increase in demand; R. Akiba and S. Nomura, “Keynote Address: Socio-Economic Considerations of Future Space Transportation Systems,” in M. Rycroft, ed., *The Space Transportation Market: Evolution or Revolution?* Kluwer Academic Publishers, 2000, p. 5. Based on SpaceX current catalog prices (SpaceX, *Capabilities & Service*, website, undated) launch service prices have fallen by a factor of four since then (to ~ \$2,500/kg). However, demand for launch services has not risen by a factor of four. In fact, if we were to remove SpaceX’s self-provisioned launches for the Starlink satellite constellation, the launch services market has remained largely flat, within normal variation, over the last ten years (see Figure 1.2). This is because launch service prices are only one of many expenses incurred in building and deploying space systems, and the uncertainty associated with those other expenses is typically much larger than the cost of launch.

¹² OneWeb has a constellation of approximately 640 satellites in LEO that supply internet-based communications across the globe. The constellation build-out was approximately 70 percent complete when the invasion of Ukraine occurred; Loren Grush, “Russia Holds OneWeb Rocket Launch Hostage, Issues Conditional Demand,” *The Verge*, March 2, 2022.

¹³ The only qualification requirement for award of a Lane 1 launch is to have had one successful prior launch for an equivalent mission. For example, a vehicle that had previously launched a similar mass to LEO would likely be considered acceptable for launch to a geosynchronous transfer orbit.

Our analysis seeks to understand how decisions made in the NSS market affect U.S. launch providers and their ability to participate in the global addressable market, as well as the attendant return impacts on the NSS market. As such, it is necessary to define the portion of the global markets that is addressable by U.S. launch providers. As was done for the 2019 assessment, we define the global addressable share by excluding those portions of the global demand that are not competed in the open market. We exclude launches where selection of the provider is the subject of international agreements, national security launches for nations that have their own domestic launch capability, launches of payloads that are closely associated with national pride for nations that have domestic launch capability, and launches where the payload manufacturer or space service provider is also a launch service provider (i.e., self-provisioning). Figure 1.2 plots the number of launches in the global

Figure 1.2. Annual Global Heavy Lift Launches



SOURCE: RAND analysis of launch type, RAND's Heavy Lift Launch Data Set, 2007–2023.

NOTE: Categories of launch are as defined in Triezenberg et al., 2020:

- **Addressable:** Sections of the launch market that are addressable by any U.S. launch provider. The addressable portion is defined by what is not fenced off.
- **International agreement:** Launches to fulfill an international obligation, such as those of the Belt and Road Initiative, the European Union (EU), or the International Space Station (ISS).
- **National affinity:** For nations with domestic launch capability, any payload that is sufficiently entwined with national pride, including launches to the moon or other planets. If a nation does not have domestic launch capability, launch of these payloads may fall into another category, including addressable.
- **National security:** For nations with domestic launch capability, the launches related to that nation's national security, such as military communication, reconnaissance, or navigation (Global Positioning System [GPS]) satellites. If a nation does not have domestic launch capability, its national security launches may fall into another category, including addressable.
- **Self-provisioning:** For vertically integrated organizations/firms that have their own launch capability, the related launch service is assumed to be nonaddressable by competitors.

heavy lift launch market over time as a function of this categorization. While the self-provisioning segment of the heavy lift launch market has seen significant growth (primarily due to SpaceX's Starlink launches), the rest of the market is flat (with normal year-to-year variation), and the share competitively addressable by a U.S. provider is less than 20 percent of the total.¹⁴

The Organization of This Report

With the above definitions as background, Chapter 2 of this report examines current and potential U.S. and international suppliers to the global heavy lift launch market. Chapter 3 examines trends that may indicate possible sources of growth in the addressable demand, summarizes the status of the NSS Phase 2 demand, and describes our projections of future demand for both the addressable and NSS heavy lift launch markets. Chapter 4 discusses how supply and demand are likely to interact over a range of possible scenarios in the next ten years and the impact on U.S. launch providers. Finally, Chapter 5 summarizes our overall findings and makes recommendations that the USSF should consider as it moves forward with the NSS launch program. In the appendix, we describe the Monte Carlo analyses that are at the heart of our assessments regarding possible future dynamics in this market over the next ten years.

¹⁴ If we exclude the self-provisioned launches, the addressable share remains steady, at 35 percent of the market. The *self-provisioned* launches are dominated by those needed to launch Starlink, a constellation of nearly 4,000 small satellites in LEO. Approximately 130 self-provisioned Falcon 9 launches in support of the constellation were made in years 2020 through 2023. In 2023, nearly half (48 percent) of all heavy lift launches were Starlink launches. Other proliferated satellite constellations (e.g., Iridium NEXT, OneWeb) are included in the *addressable* share. While these proliferated constellations have not created growth in addressable demand for launch services, they do represent a shift of communication services from their traditional location in geosynchronous earth orbit (GEO) to lower orbits. For further discussion, see Chapter 3.

Supply in the Heavy Lift Launch Market

In this chapter we explore the current state of supply in the heavy lift launch market. First, we update our list of “currently active” suppliers, note changes in their status since the 2019 assessment, and discuss the impact of those changes on both the global commercial and NSS-certified markets. Second, we provide an update on new entrants to the market. Given the distinct lack of success among the new entrants assessed in 2019, we conclude the chapter with a discussion of some of the major setbacks experienced by different entities and how approaches to product development (measured by technology readiness levels [TRLs]) interact with market development (measured by innovation readiness levels [IRLs]).

Current Launch Vehicles Competitive in the Heavy Lift Launch Market

Table 2.1 identifies currently active providers that are capable of meeting the heavy lift launch criteria in one or more of the orbits shown in Table 1.1. We define *active* as having flown at least twice in the years 2021 and 2022. Retirements and market withdrawals since the 2019 assessment are highlighted.

Table 2.1. Active Heavy Lift Launch Vehicles

Country of Manufacture	Vehicle	Launches in 2021 and 2022	Addressable Launches in 2021 and 2022	Change since 2019 assessment
United States	Antares	4	4	Retired; final launch August 2, 2023
United States	Delta IV Heavy	2	0	Retired; final launch April 9, 2024
United States	Atlas V	11	3	Retired; all vehicles sold
United States	Falcon 9	91	26	
Russia	Soyuz FG	38	11	Under sanctions
Russia	Proton M	3	1	Under sanctions
China	Long March 3 B/E	15	0	No longer active in addressable market
China	Long March 5	3	0	New entrant but not active in addressable market

Country of Manufacture	Vehicle	Addressable Launches in		Change since 2019 assessment
		2021 and 2022	2021 and 2022	
China	Long March 7	7	0	New entrant but not active in addressable market
EU	Ariane 5	6	5	Retired; final launch July 5, 2023
Japan	H-II A	2	1	In process of being retired

SOURCES: RAND analysis of data recorded in RAND's Heavy Lift Launch Data Set, 2007–2023.

For the global addressable market, the Atlas V, Ariane 5, and Delta VI Heavy launch vehicles have been retired from the market. Japan appears to be phasing out the H-II in expectation that the H-III will better serve the market. While China has two new active Long March vehicles (Long March 5 and 7) and its Long March 3 series remains active, none of the Chinese launch vehicles have participated in the addressable market in recent years; China has instead dedicated those vehicles to domestic launch. India's GSLV-III does not meet the mass-to-orbit definition for heavy lift launch and thus did not make our list of active vehicles. The GSLV-III is, however, active as a medium lift launch vehicle and could be a viable provider of launch for aggregates of payloads (such as was done by OneWeb in the immediate aftermath of its decision to terminate its contracts with Soyuz). Given sanctions on Russian technology, SpaceX's Falcon 9 may be the only viable heavy lift launch provider for many commercial buyers. This gives SpaceX a monopoly position in the addressable market, and the dearth of competition has resulted in a rash of new firms seeking to enter the market. As we documented in our earlier research, buyers in the addressable market value diversity in supply, and commercial buyers are currently actively supporting the possible new entrants with both orders and financing. Details regarding this support are provided in Chapter 4.

For the NSS-certified market, the Delta IV Heavy made its final launch on April 9, 2024, and all remaining Atlas V launches have been allocated.¹ This leaves SpaceX's Falcon 9 and Falcon Heavy as the only available NSS-certified launch vehicles. While the Falcon Heavy is rarely used in the addressable market, as of May 2023 it regained a status of "active," having made one addressable launch and two NSS launches within a two-year period.

New Entrants to the Heavy Lift Launch Market

Table 2.2 lists launch vehicles currently under development, where the supplier has indicated an intention to enter the heavy lift market per our definition of satisfying at least one of the mass-to-orbit requirements in Table 1.1. Two of the U.S. entrants, New Glenn and Vulcan, have publicly stated their intention to become NSS certified, and both were also competitors for the Phase 2 contract in 2019.² Upon NSS certification, they will be eligible for Lane 2 missions awarded under Phase 3.

¹ Sandra Erwin, "End of an Era: Delta 4 Heavy Soars One Last Time," SpaceNews, April 9, 2024.

² An additional competitor for the Phase 2 contract, Northrop Grumman's Omega vehicle, abandoned its attempt to enter the market after it was not selected. Stephen Clark, "Northrop Grumman Ends Omega Rocket Program," Spaceflight Now, September 14, 2020.

Table 2.2. New Entrants to the Heavy Lift Launch Market

Country/Firm of Manufacture	Vehicle	Expected First Launch, from 2019 assessment	Actual or Expected First Launch^a
United States/SpaceX	Super Heavy/Starship	2020	Two attempts: 2023 (failed); 2024 (failed)
United States/Blue Origin	New Glenn	2021	Mid-2024
United States/ULA	Vulcan	2021	January 2024 (success)
Russia/Roscosmos	Irtysh (Soyuz 5)	2022	2025
Russia/Khrunichev	Angara A5	—	2020 (success); 2021 (failed); 2024 (success)
EU/Arianespace	Ariane 6	2020	2024
Japan/Mitsubishi	H-III	2020	2023 (failed); no announced next attempt
United States/Relativity Space	Terran R	—	2026
United States/Rocket Lab	Neutron	—	2025
United States/Northrop Grumman	Antares 330	—	2024

NOTE: ^a Expected first launch dates are from the most recent manufacturer press releases as of November 2023; actual launch dates are as of April 2024.

However, not all new U.S. entrants have expressed an interest in NSS certification, indicating that they may be planning to compete only for Lane 1 launches under the Phase 3 contract.

Among the non-U.S. entrants, Russia plans to bring its Irtysh (Soyuz 5) rocket to the market in 2025, though its viability on the global addressable market will depend on Russia's geopolitical standings and economic sanctions.³ The Russian Angara A5 had a successful launch in 2020, a failed launch in 2021, and a successful launch on April 11, 2024.⁴ Notably, this 2024 launch used the Vostochny Cosmodrome, a relatively new space port on Russia's east coast, reducing Russia's reliance on the Baikonur Cosmodrome leased from Kazakhstan. As with the Irtysh (Soyuz 5), the viability of Angara A5 in the commercial markets will depend on geopolitical standings and economic sanctions.

The EU entrant, Ariane 6, and the Japanese entrant, H-III, are intended to replace retired/retiring heavy lift launch vehicles at lower costs and/or greater efficiencies. Ariane vehicles have long competed in the addressable markets, and the Ariane 5 was a market leader for several years. Ariane 6 was designed with the intent to compete in the addressable markets while also providing launch for

³ "First Launch of Soyuz-5 Rocket Due Dec 24, 2025," TASS, August 17, 2023.

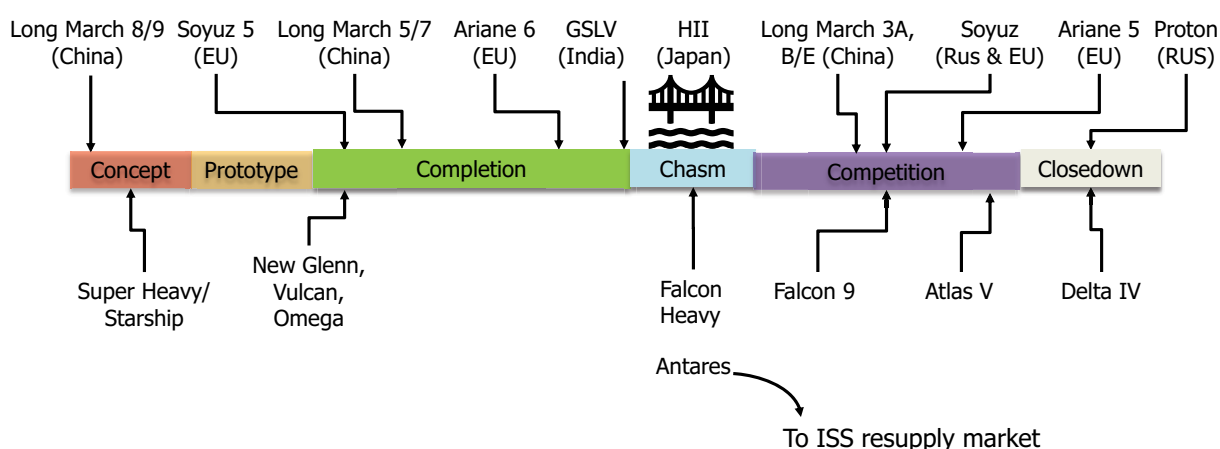
⁴ It is unclear if the April 2024 launch met the mass-to-orbit requirements for heavy lift launch. The only publicly revealed payload is an experimental 3U CubeSat into LEO (a 3U CubeSat is quite small—measuring just 30 × 10 × 10 cm, with a typical mass of 6 kg). A mass model was also launched into GEO; it remains attached to the final stage booster. We are unable to find independent verification of total payload mass. See "Angara-A5 Rocket Launched from Vostochny," Spacewatch.Global, April 12, 2024.

European national security and national affinity payloads. It is harder to judge the intent for the H-III with respect to the addressable heavy lift launch market. While Mitsubishi, the maker of the H-III, accepts contracts from commercial space service providers, it does not appear to actively market its launch services.

New Entrant Challenges

We use the innovation readiness level construct to plot the positions of current and new entrants in the global addressable market.⁵ Figure 2.1 shows the positioning as we assessed the existing and new entrant providers in 2019.⁶ Figure 2.2 shows our assessment of the providers in 2023.⁷

Figure 2.1. The Positioning of Providers in the Global Addressable Launch Market as of 2019



SOURCE: Reproduced from Triezenberg et al., 2020, Figure 3.2, p. 27.

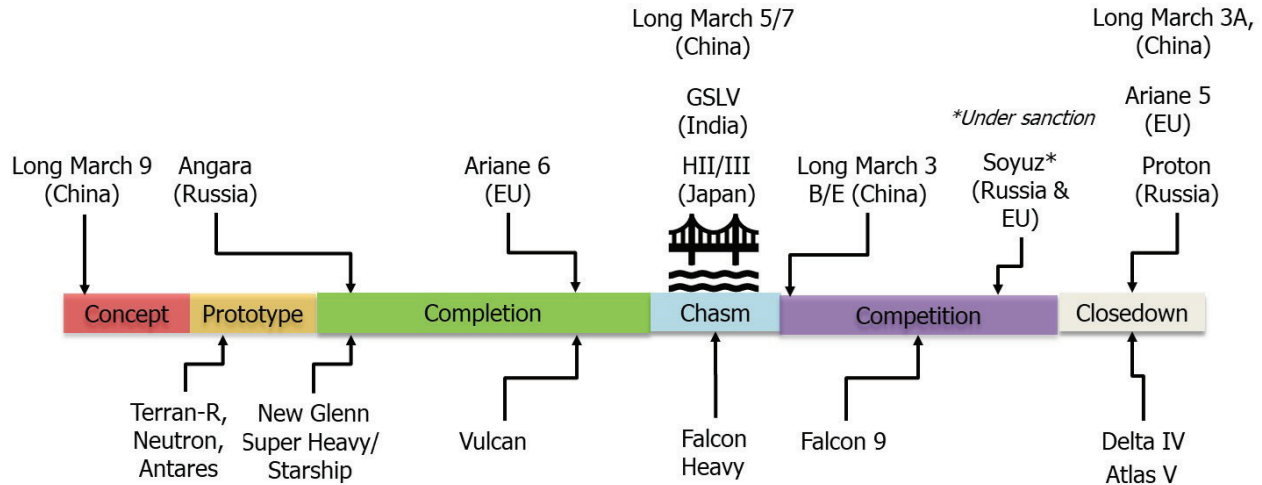
⁵ Ming-Chang Lee, To Chang, and Wen-Tien Chang Chien, "An Approach for Developing Concept of Innovation Readiness Levels," *International Journal of Managing Information Technology*, Vol. 3, No. 2, May 2011, defines the levels as follows:

- (a) Concept: Basic scientific principles of the innovation have been observed and reported, and the critical functions and/or characteristics have been confirmed through experiments (equivalent to TRL 1–3).
- (b) Prototype: Components have been developed and validated, and a prototype has been developed to demonstrate the technology (equivalent to TRL 4–6).
- (c) Completion: Technological development has been completed, and the complete system functionality has been proven in the field (equivalent to TRL 7–9)
- (d) Chasm: Chasm in the IRL framework refers to the challenges and difficulties that a new entrant or product may encounter when first introduced to the market (early stage).
- (e) Competition: This is the mature phase of the market, when it has reached a state of equilibrium marked by the absence of significant growth or innovation. The goal is to maintain and enhance the position of the product within the market and to cope with competition.
- (f) Changeover/Closedown: These are the two options in the declining stage of the market. . . . *Closedown* means the innovation has come to obsolescence and exits.

⁶ For the 2019 assessment, we placed all new entrants then competing for the Phase 2 National Security Space Launch contract (New Glenn, Omega, and Vulcan) in the same position on the IRL so as not to give the impression of preference.

⁷ The 2023 assessment was done in early fall of 2023. This was prior to the Super Heavy/Starship's second and third integrated flight tests (November 2023 and March 2024, both unsuccessful but showing progress); the Vulcan's first successful flight test (January 2024); and the Angara's first successful test flight (April 2024).

Figure 2.2. The Positioning of Providers in the Global Addressable Launch Market as of 2023



It is extremely difficult to enter the launch market, and launch vehicles rarely mature as planned. As we noted in the 2019 assessment, “Hypothetically, if all of these [new entrant] launch vehicles were to mature as planned, in 2022 there would be an astounding 19 vehicles competitive in the heavy lift launch market.”⁸ However, of the new entrants listed in 2019, only the Falcon Heavy and the Long March 5 and 7 have had a successful first launch as of December 2023 and, as shown in Table 2.1, none of those are active in the global addressable market. This demonstrates that it often takes several years before a vehicle claims a significant share of the global addressable market; the provider is unable or unwilling to cross the chasm either because the vehicle does not match market need (e.g., the Falcon Heavy) or because the provider chooses to compete only in domestic markets (e.g., the Long March family of launch vehicles).

To help readers better understand the difficulties of getting to a first launch, we use Ariane 6 as an example of how a seemingly very mature entrant (we rated it as close to completion in 2019) can stall for four years. The Ariane 6 rocket was initiated by the European Space Agency (ESA) in 2014 as a replacement for the Ariane 5 and is developed by ArianeGroup, a joint venture of Airbus and Safran and parent company to Arianespace. The goal of the program is to reduce launch costs to better compete in the addressable market and to provide European nations with an independent access to space for their national affinity and national security payloads.⁹ Political considerations among major ESA members like France, Germany, and Italy mean that production is spread across multiple countries and contractors. The development budget in 2023 is at US\$4.4 billion, nearly double original estimates.¹⁰ Initially slated to be launched in 2020, the program has faced considerable delays,

⁸ Triezenberg et al., 2020, p. 19.

⁹ ESA, “Ariane 6 Overview,” webpage, undated.

¹⁰ Eric Berger, “Europe’s Ariane 6 Rocket Is Turning into a Space Policy Disaster,” *Ars Technica*, April 18, 2023a.

including those caused by the coronavirus disease 2019 (COVID-19) pandemic, supply chain issues, and technical issues; the first launch has yet to be attempted.¹¹

In 2020, technical issues were discovered with a cryogenic arm, and in 2022, issues with the Vinci upper-stage engine required design of a new power unit. More recently, on July 18, 2023, an operational test was conducted to verify nominal interactions between the Ariane 6 vehicle and its ground equipment, including the core-stage tank filling. Although the majority of the testing objectives were accomplished, the final stage of the test, which include a short hot firing of the Vulcain 2.1 main engine, was not accomplished.¹² ESA believes that a “reconfiguration of the ground system” will be needed before this test can be performed.¹³ As of April 2024, the first launch is planned for summer 2024, but no specific date has been set. This leaves a gap in the European market, since the final launch of Ariane 5 took place in July 2023.

After nearly a decade in development, the vehicle’s capabilities and launch costs may not be competitive.¹⁴ To simplify design and ideally avoid risk, Ariane 6 is not designed for reuse, contrary to the currently popular reusable designs of launch vehicles like the Falcon 9, Starship, New Glenn, Neutron, and Terran R. This decision has led the French minister of the economy, Bruno LeMaire, to say that perhaps ESA “didn’t take the right path.”¹⁵ We note, however, that no reusable vehicle other than Falcon 9 has yet had a first successful launch, let alone demonstrated the scale required to make reusability profitable.

If Ariane can achieve a successful first launch prior to the other new entrants, it may yet gain a foothold in the market, especially in Europe. The Ariane 6 program has received a total of 29 orders, including eight Galileo navigation satellites and 18 launches from Amazon’s Project Kuiper.¹⁶ However, delays for the first launch have forced the European Commission to seek an ad hoc security agreement to launch its Galileo satellites from U.S. soil using the Falcon 9 launch vehicle.¹⁷

We close this discussion of IRLs and market positioning with a reflection on the need to build and maintain buyer enthusiasm during the lengthy period of completion needed to mature new technology (TRLs). We use two cases, New Glenn and Super Heavy/Starship, both of which were in the early stages of development when we performed our 2019 assessment, to illustrate how approaches to product development (TRLs) interact with market development (IRLs).

¹¹ Bella Richards, “Ariane 6’s First Flight to Slip into 2024,” NASA Spaceflight, May 15, 2023; Andrew Jones, “1st Launch of Europe’s New Ariane 6 Rocket Slips to 2024,” Space.com, August 9, 2023a. In fact, shortly before this report went to press in July 2024, Ariane 6 had a successful first launch.

¹² European Space Agency, “N° 38–2023: Media Invitation: Ariane 6 Test Campaign Update,” press release, August 8, 2023.

¹³ Ibid.

¹⁴ The Ariane 6 is expected to be priced between \$77 million and \$126 million depending on the number of boosters, which is significantly cheaper than the estimated \$175 million price tag of the Ariane 5; Richards, 2023. For comparison, SpaceX’s Falcon 9 is reported to be priced at \$67 million. Note that price need not reflect cost. Michael Sheetz, “SpaceX Raises Prices for Rocket Launches and Starlink Satellite Internet as Inflation Hits Raw Materials,” CNBC, March 23, 2023.

¹⁵ Bruno LeMaire, quoted in Joshua Posaner, “Europe’s Lack of Rocket ‘Audacity’ Leaves It Scrambling in the Space Race,” *Politico*, January 15, 2021.

¹⁶ Caleb Henry, “EU Reserves Four Ariane 6 Rockets for Galileo Navigation Satellites,” SpaceNews, January 23, 2020; Arianespace, “Arianespace Signs Unprecedented Contract with Amazon for 18 Ariane 6 Launches to Deploy Project Kuiper Constellation,” press release, April 5, 2022.

¹⁷ Joshua Posaner and Laurens Cerulus, “EU Turns to Elon Musk to Replace Stalled French Rocket,” *Politico*, April 17, 2023.

If a launch vehicle of the Super Heavy/Starship class of lift is to be financially successful and cross the IRL chasm, SpaceX needs to radically transform the market for launch services.¹⁸ We make this assertion based on our observation that, as of early 2024, there are few signals of viable commercial business cases enabled by super heavy lift, and hence few potential buyers. Building and maintaining enthusiasm with respect to the vehicle's potential is essential to eventually building and sustaining a market for it. SpaceX is using a "launch often and learn things" development approach to maturing the Super Heavy/Starship's TRL that has so far resulted in three unsuccessful integrated flight tests. As we documented in our 2019 discussion regarding learning from history, the launch markets do not tolerate mission failure well. However, it is our assessment that visible progress regarding technology readiness has been made with each launch. This visible progress on the TRL may create certainty regarding ultimate success, which in turn may bring needed business cases forward to mature the IRL, creating a virtuous cycle. This approach to demonstrating incremental progress of the TRL to create enthusiasm needed to mature the IRL is not without risk, however. If the next launch attempt does not show substantial TRL progress, it could be a death blow to nascent business cases.

In contrast, Blue Origin has taken a different approach to maturing the TRL for the New Glenn and has performed few public demonstrations of the technologies needed.¹⁹ In 2019, we found substantial buyer enthusiasm for the potential business cases that New Glenn could support (the size of the fairing was particularly attractive to buyers). While New Glenn still has a solid backlog of missions, it is our assessment, based on discussions with stakeholders, that buyer enthusiasm has waned. Given that the New Glenn is designed to support established and evolving business cases, rather than entirely new ways to make money, the lack of public visibility into TRL progress may not matter to the IRL. As with the Ariane, if New Glenn can be brought to market prior to the other new entrants, it is likely to gain a foothold in the market.

¹⁸ While it is difficult to ascertain the eventual performance of the Super Heavy/Starship, as of Spring 2024 SpaceX says it will be capable of lifting 100 to 150 tons into orbit; SpaceX, "Starship: Service to Earth Orbit, Moon, Mars and Beyond," webpage, undated. Although the orbit is not stated, we presume it to be LEO or polar orbit. This is over ten times the mass-to-orbit capability for heavy lift launch defined in the present report.

¹⁹ The exception to this statement is the very public successful first launch of the Vulcan, which uses a variant of the Blue Origin-developed engine (BE-4) needed for the New Glenn.

Heavy Lift Launch Demand

Our research seeks to understand how trends in the demand for heavy lift launch affect U.S. suppliers to that market. Therefore, in this chapter, we update the historical analysis of the global addressable market for heavy lift vehicles to reflect four years of change and examine the overall trends to discover possible sources of growth. We then forecast the addressable demand for the next ten years. We also provide a short recap of NSS demand, to include Phase 2 awards.

Addressable Launch Market Demand

The size of the addressable heavy lift launch market has been relatively stable over the last 16 years.¹ The average has fallen slightly since the 2019 assessment to just 19 launches per year, given relatively lackluster demand in 2019 and 2020. Demand peaked in 2022 and fell, but remained above average, in 2023. In the following sections, we plot the demand by year for a variety of categorizations and examine them for market trends.

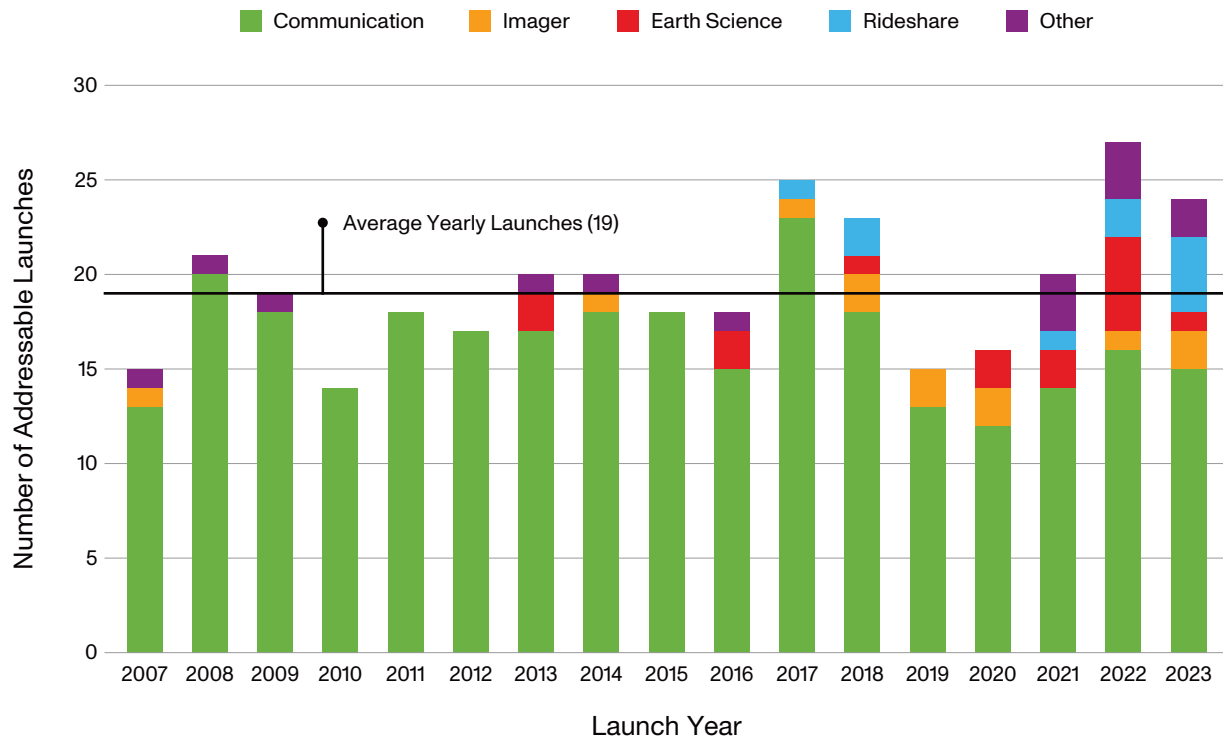
Sources of Addressable Demand

Figure 3.1 shows how addressable demand has varied from 2007 to 2023 by type of payload. As can be seen, payload type has become more diversified and is no longer the sole domain of communications satellite missions. While demand from satellite communications service providers fell and remained flat from 2019 to 2023, the demand from imaging and earth science missions grew, in part due to more nations establishing space agencies.² In cases where a nation does not have domestic launch capability, these national space agencies become a new source of demand within the addressable market.

¹ Note that the number of annual launches is, theoretically, a function of both supply and demand. However, as we discussed in the 2019 assessment, the launch market is not price sensitive, and there has been no constraint on supply since the addressable market was created in the aftermath of the *Challenger* space shuttle disaster. Therefore, we use annual launches as a proxy for demand.

² Renata Knittel Kommel, Ashley Peter; Mackenzie Puig-Hall, and Luc Riesbeck, *Exploring Insights from Emerging Space Agencies*, Center for Strategic and International Studies, October 2020.

Figure 3.1. The Global Addressable Demand Is Diversifying



SOURCE: RAND analysis of primary payload type, RAND's Heavy Lift Launch Data Set, 2007–2023.

For small- to medium-size payloads, rideshare (the simultaneous launch of multiple payloads from different customers) has become an affordable alternative to purchasing a dedicated launch vehicle and a possible source of growth for heavy lift launch providers. SpaceX offers quarterly rideshare Falcon 9 launches through its Transporter series, priced at 50 kg of payload for US\$275,000. Russia previously offered rideshare missions, but none have occurred since 2019. India's GSLV and Polar Satellite Launch Vehicle (PSLV), operating as small and medium lift launch vehicles, also compete for rideshare customers and have in the past been competitive in price.³

Among the “other” category missions, there may be growth in what we term *space commerce* and *space science*. *Space commerce* includes missions to repair or otherwise extend the life of satellites, as well as missions to explore space-based sources of production, some crewed. *Space science* includes missions to explore asteroids, the sun, and the moon. While space science missions have traditionally

³ To understand whether this growth in heavy lift rideshare might be coming at the expense of the addressable demand for small and medium lift launches, we developed a small and medium lift launch dataset for the years 2015–2023 (using the same sources and methods described in the appendix as were used to construct the heavy lift launch dataset). In examining that dataset, we found steady growth in demand for small and medium lift launch that was uncorrelated with rideshare launches. While this might be an indicator that the low price of launch for a small satellite using heavy lift rideshare is creating new demand from entities who could not otherwise afford to conduct activities in space (as opposed to simply showing a shift in how small satellites are packaged), we caution that our statistics are global. The growth we see in the small to medium lift addressable market is driven primarily by China's domestic commercial satellite businesses, and it may be that heavy lift rideshare is indeed suppressing growth for small and medium lift launch vehicles from nations outside China. That is, outside China it is likely that there has been a shift in how satellites are packaged—creating growth for heavy lift launch—but not growth in overall demand for launch services.

been confined to nations with domestic launch capability and thus categorized as national affinity launches, nations without domestic launch capability have begun participating in space science and are purchasing launch services in the addressable market.

Before leaving this discussion on sources of demand in the global addressable market, we address the question of why the advent of proliferated satellite constellations has not led to an increase in the addressable demand for launch services. As noted in the 2019 assessment, “demand from proliferated constellations of small satellites is expected to be small,” and the business cases were yet to be tested.⁴ In the four years since, business cases have been tested and all potential constellations have scaled back or delayed their plans. Current plans are described in Table 3.1.

Table 3.1. The Status of Proliferated Satellite Constellations

Company	Orbit	Number of Satellites	Satellites per Launch	Launches	Launch Provider	Planned Initial Service Date
OneWeb	LEO	648	~30	17.5 (24 predicted)	Falcon 9, GSLV, Soyuz ^a	Actual date, May 2022 (2021 predicted)
	MEO	<i>Less than 1000</i>		<i>Unknown</i>		<i>Unknown</i>
Telesat	LEO	117	~10	12	PSLV, Soyuz	Mid-2026 (was 2022) ^b
SpaceX Starlink	LEO	4,409 (12000 predicted)	~ 40 for v1 satellites, ~ 20 for v2 satellites	127 (74 predicted; larger number is due to satellite weight growth)	Self-provisioned	1.5M subscribers reported as of May 2023, far below planning ^c
	Very low earth orbit	<i>Perhaps abandoned</i>		<i>126 predicted</i>		<i>Unknown</i>
Amazon Kuiper	LEO	578 in Phase 1 (of 5)	~35	91	Ariane 6, Atlas V, New Glenn, Vulcan ^d	Fourth quarter, 2024 (was 2020)

NOTE: Immature plans are shown in *italics*.

^a OneWeb launches occurred between February 2019 and March 2023. OneWeb initially contracted with Soyuz for launch services. After Russia’s invasion of Ukraine, OneWeb switched to SpaceX’s Falcon 9 and India’s medium lift GSLV to complete deployment of the constellation.

^b Telesat has launched three test satellites. The first was launched on a Soyuz launch vehicle on November 28, 2017, but was lost due to an issue with the upper stage. The second was successfully launched on January 12, 2018, on a medium lift Indian PSLV rideshare. The third demonstrator was launched on July 17, 2023.

^c Viktor Hendelmann, “Starlink Surpasses 1.5 Million Subscribers, Experiences Temporary Outage,” *Starlink-Insider*, May 2023.

^d Amazon Kuiper has placed launch contracts for ULA’s Atlas V and Vulcan, and for the Ariane 6, but has yet to launch any test satellites. It also plans to self-provision using Blue Origin’s New Glenn. First launches are planned for 2024.

⁴ Triezenberg et al., 2020, p. 31.

Over the last four years, OneWeb and Starlink completed build-out of their initial constellations, and existing smaller constellations, such as Iridium NEXT and O3b, began their replenishment launches. OneWeb survived a period of bankruptcy when its business case failed and is now partially owned by the UK government. Fewer OneWeb launches were made than originally predicted, as the firm reduced constellation size and scaled back objectives.⁵ Starlink has also scaled back plans (including plans for a constellation in very low earth orbit) and has begun launching its heavier second-generation Starlink satellites. These heavier satellites reduce the number of satellites that will fit on a Falcon 9 and has increased the number of launches required to deploy the constellation, increasing total investment cost.⁶ To support replenishment and sustainment of the Starlink constellation while also supporting NSS and addressable demand, SpaceX has increased its Falcon 9 launch cadence to once a week, with a goal of 100 launches per year.⁷ However, a lack of subscribers, heavier satellites, and more launches all negatively affect Starlink's business viability.⁸

We do not anticipate any constellations arising within the next five years to replace OneWeb's addressable demand for launch services. Amazon Kuiper and Telesat, two other proposed proliferated LEO constellations, had nebulous late-2020 initial service dates. Telesat has one functional prototype satellite in orbit, while Amazon Kuiper is still a concept.⁹ It is uncertain how sustainable proliferated LEO constellations are, and this market might collapse if the core business case of supplying high speed/low latency internet service from space is weaker than firms' scaled-back plans have accounted for. As we noted in the 2019 assessment, the difference between the number of "underserved" internet users and the number of people willing to pay for high-quality space-based internet service may be much larger than these business models anticipate.¹⁰ Higher interest rates for capital may also adversely affect business models. Based on low subscription and adoption rates to date, the underlying satellite communications market may not have sufficient demand to generate the revenues needed to sustain the existing constellations.

One final question to address before moving on is whether GEO is still a preferred destination for satellites given the growth of proliferated LEO constellations. Figure 3.2 shows how demand has varied by destination orbit and clearly shows that destinations have diversified. However, major providers of satellite communication services (Eutelsat, Inmarsat, Intelsat, and Viasat) continue to launch new satellites into GEO, committing their resources to this domain. As the relatively small subscriber base for Starlink services demonstrates, we have not yet witnessed a mass exodus of GEO-based communications services in response to proliferated LEO alternatives.

⁵ Peggy Hollinger, "OneWeb Paid Almost \$50M in Investor's Fees for Bankruptcy Deal," *Financial Times*, November 15, 2021.

⁶ Second-generation Starlink satellites are three times the weight of first-generation satellites; Stephen Clark, "SpaceX Unveils First Batch of Larger Upgraded Starlink Satellites," *Spaceflight Now*, February 26, 2023a. SpaceX's satellites expressly designed to support U.S. government missions, called Starshield satellites, are expected to be heavier still due to increased security and resilience requirements; Brett Tingley, "SpaceX Wins \$70 Million Space Force Contract for Starshield Military Satellites," *Space.com*, October 2, 2023.

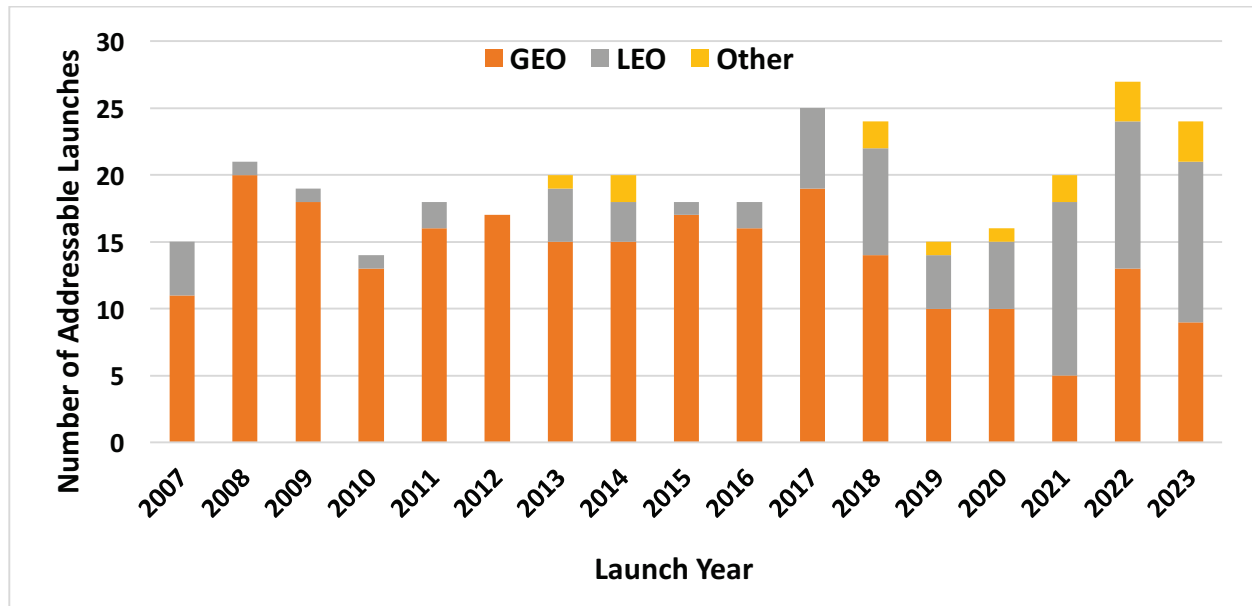
⁷ Mike Wall, "SpaceX Could Launch 100 Rockets in 2023, Elon Musk Says," *Space.com*, August 31, 2022.

⁸ Micah Maidenberger and Rolfe Winkler, "Starlink Surges but Is Still Far Short of SpaceX's Goals, Documents Show," *Wall Street Journal*, September 13, 2023.

⁹ Jason Rainbow, "Telesat Q&A | Getting Back to Lightspeed," *SpaceNews*, September 11, 2023

¹⁰ Triezenberg et al., 2020, p. 34.

Figure 3.2. Orbital Destination for the Addressable Market Is Diversifying



SOURCE: RAND analysis of orbital destination, RAND's Heavy Lift Launch Data Set, 2007–2023.

A Market Becomes a Monopoly: 2019–2022

As we noted in the 2019 assessment, there are strong buyer incentives to maintain competition in the addressable heavy lift launch market, and those market forces were not expected to allow SpaceX's Falcon 9 to continue to take a 50-percent market share. In fact, SpaceX's share dropped significantly in 2019 and remained down through 2021. The bulk of the shift in market share was to the Soyuz launch vehicle, which was selected to be a primary provider for the OneWeb constellation, in a deal brokered by Arianespace. This may have continued into 2022 had Russia not invaded Ukraine. As shown in Figure 3.3, the Soyuz addressable market share disappeared almost overnight with the invasion of Ukraine in February 2022, going from nine addressable launches in 2021 (all but one of which were from OneWeb) to two launches in 2022 (for Angola and Iran, two of the nations who do not recognize the economic sanctions imposed in the wake of the invasion).¹¹ It is unclear if and when Russian launch vehicles will return to the global addressable launch market. Prior to February 2022, Russia hoped to return to full competition with its next-generation launch vehicle, the Irtysh (Soyuz 5). As of August 2023, Russia plans to debut the Soyuz 5 on December 24, 2025.¹² While in the past Ariane 5 held a significant share of the addressable market, Arianespace launched its final Ariane 5 rockets in July 2023, bringing a conclusion to its 27-year run.¹³ Delays have pushed the maiden launch of its next-generation model, Ariane 6, to 2024.¹⁴

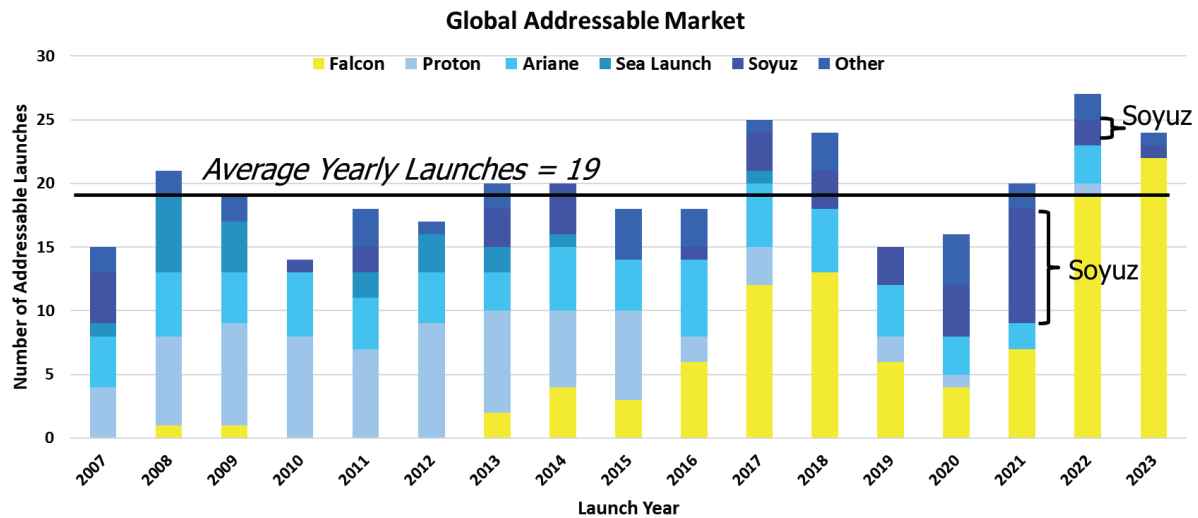
¹¹ Russia confiscated 36 OneWeb satellites awaiting launch—satellites that OneWeb has yet to see returned. Dan Swinhoe, "OneWeb Has Given Up on Retrieving Satellites Left in Russia," Data Center Dynamics, March 20, 2023.

¹² "First Launch of Soyuz-5" August 17, 2023. TASS, August 17, 2023.

¹³ Joey Roulette and Tim Hephner, "Ariane 5 Launches Final Mission as Europe Faces Space Gap," Reuters, July 6, 2023.

¹⁴ Jeff Foust, "Ariane 6 Completes Short-Duration Engine Test," SpaceNews, September 6, 2023b.

Figure 3.3. Market Share in the Global Addressable Market



SOURCE: RAND analysis of launch service provider, RAND Heavy Lift Launch Data Set, 2007–2023.

The 2019 assessment postulated China’s and India’s growth in the global addressable market and suggested that their shares would increase with an anticipated growth in launches from low- and middle-income nations. Our findings indicate that such shifts did not occur. China did increase its production and launch cadence, but this supply increase was consumed by its domestic demand.¹⁵ Western-aligned nations have generally avoided launching from China due to security and intellectual property concerns, and expected growth in demand from lower- and middle-income nations failed to materialize.¹⁶ Instead, demand from these nations was satisfied by rideshare launches, most prominently SpaceX’s Transporter series. In offering these rideshares, SpaceX appears to have consolidated market share that might otherwise have gone to China or India. India’s new entrant launch vehicle, the GSLV-III, has not kept pace with competitors for mass-to-orbit capability, and discussions with industry leaders reveal an underlying distrust of India’s launch capacity and schedule reliability and consistency, with reports of foreign-purchased flights being delayed to prioritize domestic launches. Furthermore, India failed to capitalize on Russia’s exit from the global addressable market. While OneWeb did contract with India for GSLV launches, the majority of the OneWeb post-Soyuz launches were accomplished using SpaceX’s Falcon 9.

¹⁵ Due to the extensive involvement of the Chinese government in its commercial companies, it is hard to conclusively determine which launches are truly addressable versus which involve national affinity. China does have a growing commercial space industry, producing commercial satellites and launch vehicles, but this demand is primarily satisfied using small and medium lift launches or rideshare on heavy lift launches. Most of the Long March heavy lift demand is for national affinity or national security launches. For a breakdown of China’s commercial satellite industry, see Neel V. Patel, “China’s Surging Private Space Industry Is Out to Challenge the US,” *MIT Technology Review*, January 21, 2021; and Blaine Curcio, “A Rising Chinese Space Sector: Expectations vs. Reality,” *Satellite Markets & Research*, June 1, 2022.

¹⁶ Demand from lower- and middle-income nations, as defined by the World Bank, has seen a return to the levels of the early 2010s. Where high-income countries rebounded from their 2020 pandemic-induced slump with larger-than-average numbers of launches in 2021 and 2022, the demand from middle- and lower-income nations did not rebound. Launch numbers decreased from an average of seven launches a year between 2015 and 2018 to four launches a year between 2019 and 2022. National income designations are developed by the World Bank and updated annually. For more information and an interactive tool, see World Bank, “Classifying Countries by Income,” September 9, 2021.

Factors Commercial Buyers Consider When Selecting a Launch Service Provider

As noted above, the communication service market remains the largest commercial buyer of heavy lift launches. The business model is characterized by very large up-front capital investments followed by revenue streams lasting the lifetime of the satellites and generally has a distinct multiyear capital expenditure cycle. Proliferated LEO constellations to date have had larger capital expenditures than those needed for the more traditional GEO constellations. While only a few of the proliferated constellations have begun replenishment, sustaining a proliferated constellation is anticipated to require more continuous capital expenditures and a less cyclical launch cadence than historic trends. Currently, we see no evidence of a change in the cyclical nature of demand for heavy lift launch and, for this reason, have included a cyclical component in our demand forecast.

Whether the launch destination is GEO or LEO, we assess that commercial buyers will continue to prioritize three general criteria when selecting a launch provider: reliability/insurability, scheduling flexibility, and launch vehicle lift/performance.¹⁷ Commercial buyers value reliable, flexible, and targeted launch vehicles and are “willing to pay a premium for improvements in these three areas.”¹⁸ The last four years have only heightened this willingness, and launch orders indicate that buyers are highly motivated to keep suppliers in (or help a second launch provider enter) the addressable market to provide greater assurance of scheduling flexibility. As we noted in the 2019 assessment, being certified as an NSS-certified launch provider can affect these areas and affect a provider’s ability to succeed on the addressable markets. However, while NSS certification may be a testament to a launch vehicle’s reliability, it may also force a launch manufacturer into designing a launch vehicle that is suboptimal for the addressable market.

The RAND Team’s Forecast of Global Addressable Demand

The 2019 assessment’s forecast for global demand addressable by U.S. firms was generally accurate and avoided being optimistic or pessimistic. However, impacts from the 2020 global pandemic could not have been anticipated. As shown in Figure 3.4, the number of launches in 2020 and 2021 were at the low end of our uncertainty bounds as payloads were grounded in response to the global shutdown.¹⁹ The number of launches rebounded in 2022 and slightly exceeded the predicted uncertainty bounds as payloads delayed due to the pandemic were able to launch. While 2023 was an above-average year for launches, it falls within the 2019 forecast’s uncertainty bounds.

We have updated the forecast based on developments in the global addressable market through year end 2022.²⁰ This new forecast is shown in Figure 3.5. As before, the forecast for demand in the

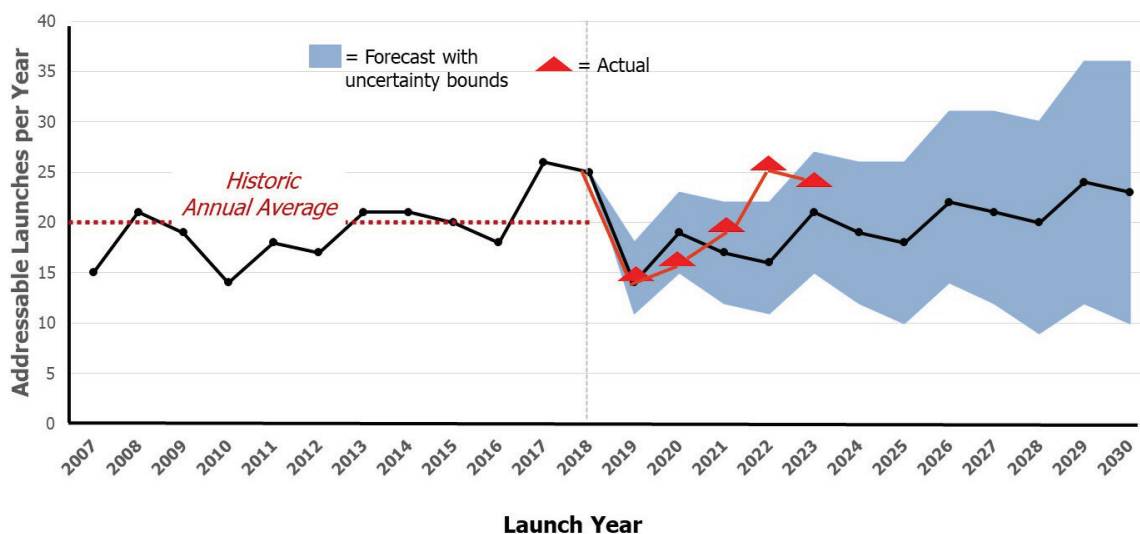
¹⁷ We characterize these criteria as being the same questions we all ask when considering transportation options: Will I arrive safely? Will I be able to leave on my schedule? And, will I be near where I need to go at the end of the ride?

¹⁸ Triezenberg et al., 2020, p. 38.

¹⁹ Integration and final checkout of the launch vehicle and payload is typically performed at the launch site. Travel prohibitions due to COVID-19 meant that launch teams could not travel to the launch site to perform these activities.

²⁰ Although we have updated the historic launch data through year end 2023 prior to publishing this report, all analytics were performed in mid-2023 using year-end 2022 data.

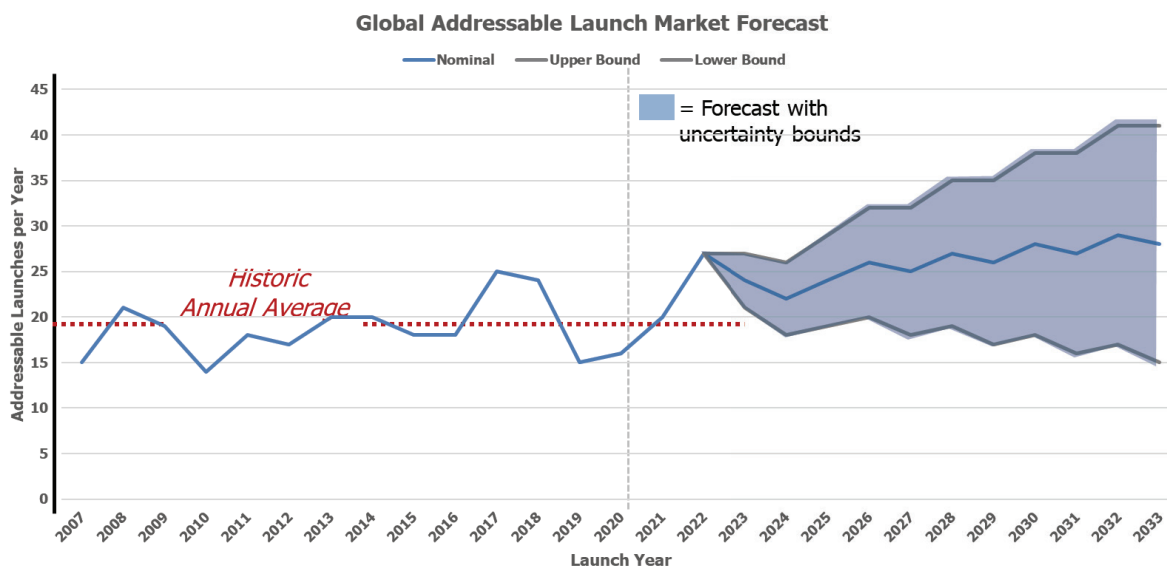
Figure 3.4. The RAND Team's 2019 Forecast Versus Actual Addressable Heavy Lift Launches



SOURCE: Updated from Triezenberg et al., 2020, Figure 3.8, p. 37.

NOTE: Actual annual launches, with the exception of those in 2022, fell within the uncertainty bounds. The number of launches in 2022 was artificially high, as launches delayed due to the COVID-19 pandemic got off the ground. While 2023 was an above-average year, it fell within our 2019 uncertainty bounds.

Figure 3.5. The Global Addressable Heavy Lift Launch Market, 2007–2022, and the RAND Team Forecast, 2023–2033



heavy lift launch market assumes cyclical demand. However, given that demand from lower- and middle-income nations failed to materialize, we assume a lower growth rate than was assumed in 2019. The growth rate in our forecast is now based on an extrapolation of the growth in earth and space science missions over the past four years. While that growth is insufficient to overcome the average decline in launches seen in the past four years, we assess that it represents a valid and growing

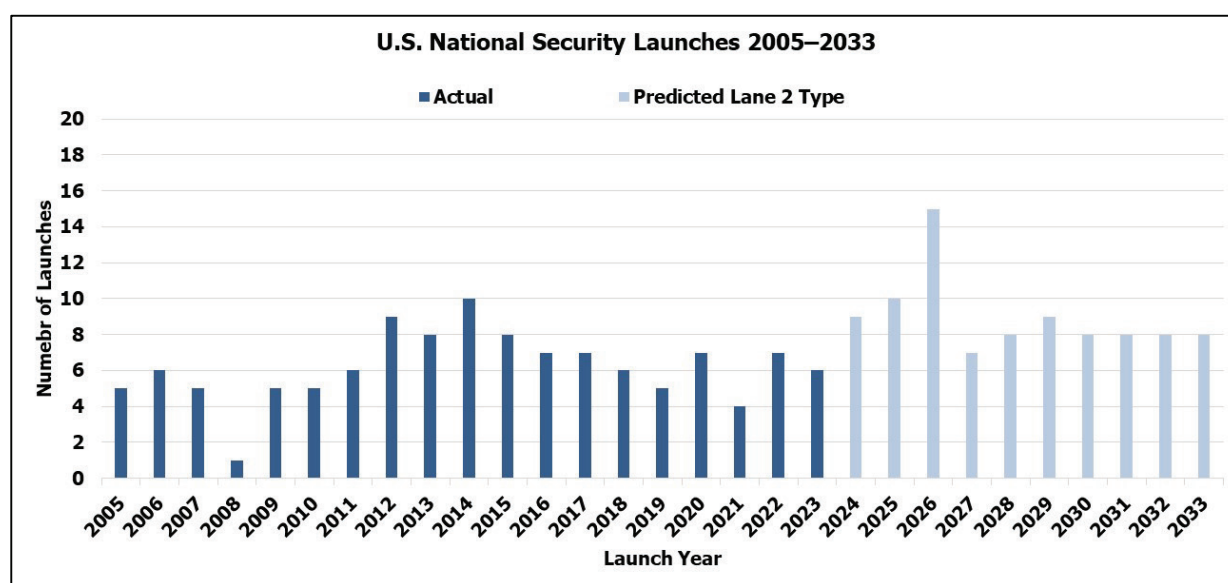
source of demand. The forecast’s uncertainty bounds are calculated based on the standard deviation of the historical demand, which increases as the forecast progresses in time.²¹

National Security Space Launch Market Demand

Historic and Predicted NSS Demand

Figure 3.6 shows actual NSS demand for launch of high-assurance payloads (i.e., Lane 2–Type launches) from 2005 to the present, and the RAND team’s estimates from the present to 2033.²² For those years covered by the Phase 3 contracts, our estimate is based on 70 percent of the Lane 2 demand projected in the Phase 3 Industry Day presentation. The discount factor of 70 percent is

Figure 3.6. NSS Demand for Launches of High Assurance (Lane 2–Type) Payloads, 2005–2023, and the RAND Team Forecast, 2024–2033



SOURCES: Data for 2005 and 2006 are taken from Boeing (Delta) and Lockheed Martin (Atlas) launch manifests, as well as Forrest McCartney, Peter A. Wilson, Lyle Bien, Thor Hogan, Leslie Lewis, Chet Whitehair, Delma Freeman, T. K. Mattingly, Robert Larned, David S. Ortiz, William A. Williams, Charles J. Bushman, and Jimmey Morrell, *National Security Space Launch Report*, RAND, MG-503-OSD, 2006. The 2007–2023 data are from RAND’s Heavy Lift Launch Data Set, 2007–2023.

NOTE: Predictions for 2024–2026 are based on recent launch manifests and NSS Phase 2 procurements. Predictions for 2027–2031 are based on 70 percent of USSF projections in the Phase 3 Industry Day presentation. Predictions for 2032 and 2033 project flat demand.

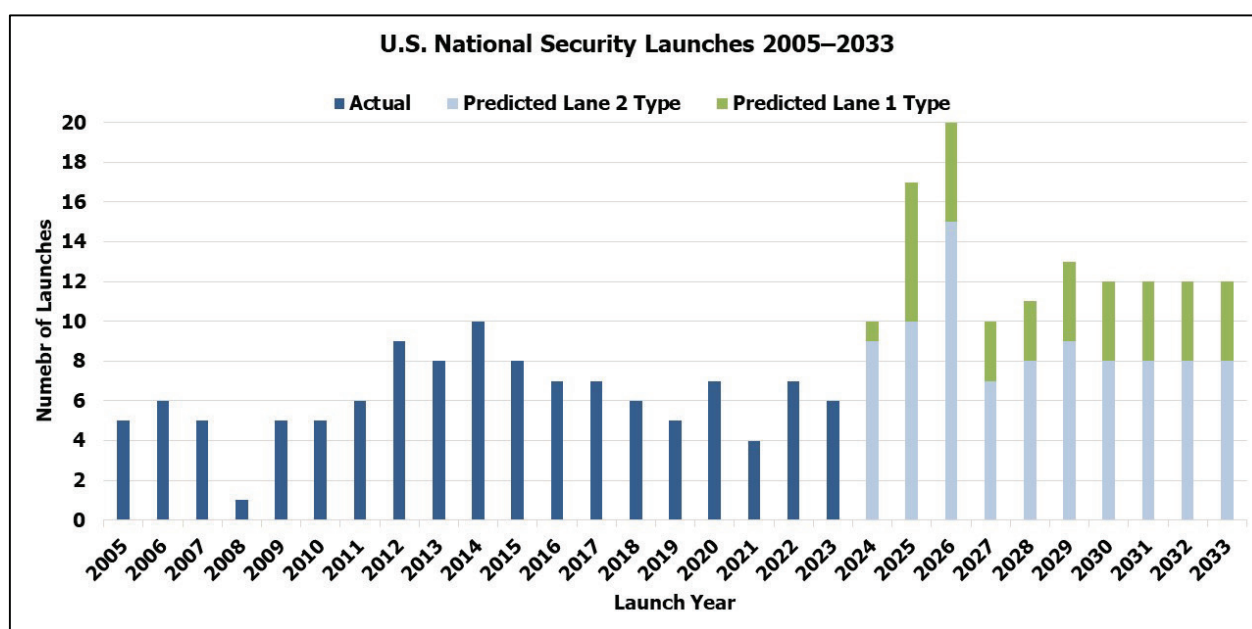
²¹ See the appendix for a more complete discussion of how the forecast and uncertainty bounds are set. Note that the number of actual 2023 launches was 24, inside our uncertainty forecast.

²² As with the addressable market, we use annual launches as a proxy for demand. In the historic data, the only year for which NSS supply was constrained was 2008. Those launches were purchased in 2006, the year of the ULA merger. The story of how the NSS market became supply constrained and the events that led to the ULA merger are discussed in some depth in the prior report in Appendix C: Learning from History; see Triezenberg et al., 2020, pp. 97–114.

based on our analysis of NSS demand projections for high-assurance missions over the past two decades. While there is a significant year-to-year variation in the forward estimates, historically this variation has been smoothed by year-to-year slips caused by either satellites currently in orbit outliving their projected design life or delays in development of payloads. Overall, NSS demand for launches of high-assurance payloads has been—and we expect it to continue to be—relatively flat.

Figure 3.7, however, adds NSS demand for launches of more risk-tolerant payloads (i.e., Lane 1–Type launches) and shows a significant burst of demand for heavy lift launch in 2025 and 2026. This burst is driven by near-term experiments and prototypes the USSF is funding and developing as it seeks to adjust its forces, strategies, and tactics to prevail in conflict with a near-peer adversary. Since these payloads are experimental in nature and their development is more directly under USSF control, we deem them less likely to be delayed or canceled.²³ Therefore, we do not apply a discount factor to Lane 1 demand projected in the Phase 3 Industry Day presentation.

Figure 3.7. NSS Demand, All Payload Types, 2005–2023, and the RAND Team Forecast, 2024–2033



SOURCES: Data for 2005 and 2006 are taken from Boeing (Delta) and Lockheed Martin (Atlas) launch manifests, as well as McCartney et al., 2006. The 2007–2023 data are from RAND’s Heavy Lift Launch Data Set, 2007–2023.

NOTE: Data for Lane 1–Type predictions are, for 2024–2026, based on recent launch manifests and NSS Phase 2 procurements; for 2027–2031 they are based on 100 percent of USSF projections for Lane 1 in the Phase 3 Industry Day presentation. Predictions for 2032 and 2033 project flat demand.

²³ The high-assurance payloads that make up the Lane 2–Type demand support joint air, naval, and land warfighting operations across the globe. As such, the USSF must negotiate the required capabilities for these payloads across the joint force and assure performance in multiple possible theaters of war, each having different stressing requirements. Satisfying those needs can result in other concerns being prioritized above delivery schedule concerns and can significantly delay payload development. In contrast, the experimental Lane 1–Type payloads have fewer stakeholders and objectives. Furthermore, by definition, the value of an experiment is in how quickly it can be carried out, allowing program offices to prioritize schedule over other concerns.

The Status of NSS Launch Contract Awards

The NSS Phase 2 procurement competitively awarded firm, fixed-price, indefinite delivery requirements contracts to two providers, SpaceX and ULA. The ordering period was five years (FY 2020–FY 2024), with a 60/40 split between the launch providers, with ULA selected for the 60-percent share and Space X selected for the 40-percent share. The projected NSS manifest was approximately 34 missions over the five-year period.

As of February 2024, the USSF has completed the Phase 2 awards. Nominally, awards are made two years before expected launch. In FY 2020, three missions were awarded, with two to ULA and one to SpaceX. Despite nominal launch dates in 2022, only two have launched. USSF-67 was launched on a Falcon Heavy in January 2023. USSF-51, the ULA mission, was originally planned as the first Vulcan NSS launch. It was later moved to an Atlas V vehicle, which is allowed under the terms of the Phase 2 contract, and launched on July 30, 2024. The third launch is for USSF-106 and is now scheduled for late 2024 as the first NSS Vulcan launch, assuming certification requirements have been completed by then.²⁴ In FY 2021, four missions were awarded: two to ULA’s Vulcan and two to SpaceX. None of these has launched. However, the Space Development Agency bought two additional SpaceX launches in FY 2021 outside the NSS contract for its Tranche 0 Lane 1–Type payloads; both launches were completed successfully in 2023.²⁵ In FY 2022, eight missions were awarded for launch in 2024 or later: five to ULA and three to SpaceX; as of August 2024, two have launched. The FY 2023 award was 12 missions, split evenly between the firms. The FY 2024 award is 21 missions, with 11 to ULA and ten to SpaceX.

Of the 48 Phase 2 missions awarded, 14 are equivalent in risk to those defined as Lane 1—that is, relatively risk-tolerant payloads designed for rapid acquisition and deployment. These missions would be poorly served if they are forced to wait for certification of the Vulcan launch vehicle, and all but three were awarded to the already NSS-certified SpaceX Falcon 9. Of the remaining 34 missions, 21 were awarded to ULA. These awards were made even though Vulcan had yet to demonstrate a successful first launch at the time of the awards. A successful first launch was accomplished in January 2024. A second successful launch is needed to complete NSS certification activities, and this is anticipated to occur mid-September 2024. The implications of this acquisition decision are a major topic of discussion in Chapter 4.

²⁴ ULA has announced their intent to launch a dummy payload as the second launch required for NSS certification in September of 2024. A successful second launch would then pave the way for USSF-106 to be launched in late 2024. Stephen Clark, “A Mid-September Test Flight of Vulcan Could Permit a Military Launch This Year,” *Ars Technica*, July 22, 2024.

²⁵ We count these two Tranche 0 launches as U.S. national security (i.e., NSS) launches in our database, even though they were bought independently. All future launches for the Space Development Agency Tranche 1 and 2 payloads have been bought under the Phase 2 NSS contract and make up the bulk of Phase 2 Lane 1–Type missions we defined for the analysis documented in this report.

Assessing the Impact of NSS Launch Acquisition Decisions

This chapter examines the ramifications of the Phase 2 launch acquisition decision, as well as more recent decisions regarding the Phase 3 NSS launch acquisition strategy on U.S. launch providers, on the NSS market, and on the global addressable market. We close the chapter with a discussion of factors that may affect the future resilience of NSS launches.

The Impact of NSS Acquisition Decisions on U.S. Launch Providers

In this section, we discuss each of the current and potential U.S. heavy lift launch providers and the changes in its business case or market position over the past four years. Where we assess that USSF decisions regarding the NSS market have affected a provider, we highlight that impact.

SpaceX

As we documented in Chapter 2, SpaceX's Falcon 9 currently has a near monopoly position in the global addressable markets. Three significant events regarding SpaceX's participation in the NSS market occurred in 2019 and 2020. First, in 2019, the Falcon Heavy was certified for NSS launch, though additional work was needed for it to qualify for the Phase 2 procurement.¹ Second, in 2020, the USSF allowed SpaceX to use previously flown boosters from a GPS 4 mission for the GPS 5 mission (launches for GPS 4 and 5 were awarded under the Phase 1A NSS contract), presumably giving an NSS stamp of approval to SpaceX's reuse concept.² While reusability of the first-stage boosters and the payload fairing is estimated to significantly lower launch costs, we caution that these estimates cannot be independently verified. Furthermore, under current market conditions, there is little incentive for SpaceX to pass along cost savings to buyers.³ The third event, also in 2020, was the

¹ Sandra Erwin, "Air Force Certified Falcon Heavy for National Security Launch but More Work Needed to Meet Required Orbits," SpaceNews, September 21, 2019.

² Michael Sheetz, "Space Force Clears SpaceX to Launch Reused Rockets for Military Missions," CNBC, June 14, 2021.

³ Some estimate that this reuse may reduce SpaceX's cost for a Falcon 9 to as low as \$30 million per launch, but actual cost data are not publicly available. Inspection and refurbishment are labor-intensive activities, and the cost of skilled labor is likely a significant cost driver. See Stephen Clark, "SpaceX Is Stretching the Lifetime of Its Reusable Falcon 9 Boosters," Ars Technica, July 10, 2023b.

inclusion of funds in the first year of the Phase 2 contract for infrastructure upgrades not associated with a specific launch. SpaceX used those funds to upgrade its launch infrastructure at Vandenberg Space Force Base to support Falcon Heavy launches and to build a vertical integration facility at Kennedy Space Center. Including the infrastructure upgrades, the total funding for SpaceX in the first contract year was \$316M.⁴

We find little evidence that NSS certification of the Falcon Heavy or these infrastructure improvements have influenced the decisions of commercial buyers. There is little demand for Falcon Heavy in the addressable market, and we observe no uptick in orders. We also find little evidence that the NSS stamp of approval on reuse of the first-stage boosters has affected the addressable market, as there was no stated opposition to the practice from commercial buyers prior to this approval. NSS is a follower, not a leader, in the acceptance of reusable technology. SpaceX's dominant addressable market share, as we discussed in Chapter 3, is driven by Russian decisions more than by NSS decisions.⁵ Commercial buyers actively continue to seek out new entrants from both international and U.S. firms in efforts to break or reduce SpaceX's current monopoly position. It is our assessment that the NSS Phase 3 acquisition strategy, and especially the creation of unrestricted competition for the launch of Lane 1 payloads, is supportive of that goal and is likely to boost the prospects of new U.S.-based entrants to the heavy lift launch markets.

United Launch Alliance

While ULA won the 60-percent share of the Phase 2 launches, the solicitation required that provided launch vehicles “transition off non-allied propulsion systems.”⁶ This requirement is responsive to legislation passed after Russia annexed Crimea in 2014 that bars the Department of Defense (DoD) from procuring NSS vehicles that include Russian technology after 2022. The requirement meant that the Atlas V, with its current Russian-built RD-180 engines—for decades a highly reliable NSS launch vehicle—could not be proposed for Phase 2. The solicitation did allow for use of the Atlas V (or another proven launch vehicle) in the first two years of the contract should ULA experience delays in the development of the Vulcan launch vehicle. One NSS Phase 2 mission was moved to an Atlas V under that provision, and the last Atlas V NSS mission (USSF-51) launched July 30, 2024.⁷

While the NSS prohibition against the use of Russian technology does not affect Atlas V usage in the global addressable market, ULA has since halted production of new Atlas V vehicles, and the

⁴ Sandra Erwin, “SpaceX Explains Why the U.S. Space Force Is Paying \$316 Million for a Single Launch,” SpaceNews, November 9, 2020.

⁵ One could argue that U.S.-led sanctions against Russian launch services have had an impact on the addressable market. However, it is our assessment that the decision to hold the OneWeb satellites hostage is the more immediate reason that commercial buyers abandoned Russian launch vehicles.

⁶ U.S. Air Force, National Security Space Launch (NSSL) Phase 2 Launch Service Procurement (LSP) Request for Proposal (RFP), Solicitation FA8811-19-R-0002, May 3, 2019.

⁷ Will Robinson-Smith, “Most Powerful Atlas 5 Rocket Launches National Security Mission,” Spaceflight Now, September 10, 2023.

remaining vehicles have all been sold.⁸ Considerations as to whether and when to retire production of a launch vehicle are complex. We assess that the decision to prohibit Russian technology for NSS launches hastened the retirement of the Atlas V for other uses, but a decades-long dearth of addressable launches, as documented in Figure 4.2 of the 2019 assessment, was likely a more influential factor.⁹ ULA's nearly single-minded pursuit of the U.S. nonaddressable heavy lift launch market since its formation in late 2006 meant that ULA would have had to devote significant resources to market and maintain the viability of the Atlas V in the addressable market. This has changed in the past two years; SpaceX's monopoly has caused commercial buyers to seek out the remaining Atlas V vehicles, but that event could not have been foreseen when the retirement decision was made.

ULA is in the process of developing and certifying its next-generation launch vehicle, the Vulcan. As noted earlier, at the time of the Phase 2 awards, Vulcan's first launch was forecast to be in 2021; its first launch occurred on January 8, 2024.¹⁰ Recent issues encountered during development include a Centaur upper-stage hydrogen leak in March 2023 that caused an explosion during a test and a BE-4 first-stage engine explosion (also during a test) in June 2023.¹¹ Despite delays, buyers continue to place orders for Vulcan launch services. ULA reported in November 2023 that the Vulcan backlog has risen to "about 70 launches," the bulk of which are launches for NSS and Amazon Kuiper.¹² ULA is also reported to be for sale.¹³

Blue Origin

Blue Origin competed but was not selected for NSS Phase 2 and has yet to attempt a first launch of its New Glenn heavy lift launch vehicle. At the time of the Phase 2 acquisition, the predicted first launch date was 2021, and the most recent prediction is late 2024.¹⁴ In the 2019 assessment, the New Glenn

⁸ Buyers of the remaining Atlas V vehicles include Amazon Kuiper and Boeing's CST-100 Starliner spacecraft. See Jeff Foust, "Starliner Launches to Remain on Atlas 5," SpaceNews, May 22, 2022a; and Joey Roulette, "ULA Stops Selling Its Centerpiece Atlas V, Setting Path for the Rocket's Retirement," The Verge, August 26, 2021.

⁹ Triezenberg et al., 2020, p. 48.

¹⁰ Delays are not an uncommon occurrence for new development of launch vehicles. Our analysis assumes a uniform probability of delay between one and four years for all new entrants. As noted in Triezenberg et al., 2020, p. 51, only when first launch predictions are given in months instead of years is there a high probability of actual first launch within the next year. The December 24, 2023, prediction was made by ULA in press announcements in late October 2023; Jeff Foust, "ULA Sets Christmas Eve Launch Date for First Vulcan Centaur," SpaceNews, October 24, 2023d. In early December, the launch was moved to January 8, 2024. Jeff Foust, "First Vulcan Launch Likely to Slip to January," SpaceNews, December 10, 2023f.

¹¹ Jon Kelvey, "United Launch Alliance Creates Fix for Centaur V Hydrogen Leak, Targets Late 2023 for Inaugural Vulcan Centaur Launch," SpaceRef, July 13, 2023.

¹² Rachel Jewett, "ULA Previews Christmas Eve Vulcan Debut and the Rocket's 70-Mission Backlog," Via Satellite, November 15, 2023.

¹³ Eric Berger, "Sale of United Launch Alliance Is Nearing Its End, with Three Potential Buyers," Ars Technica, November 14, 2023c.

¹⁴ Harry Stranger, "Blue Origin Works Towards New Glenn Debut, Ramps BE-4 Deliveries," NASA Space Flight.com, June 4, 2024.

and Vulcan were rated as being roughly equal in immaturity. One reason for a lack of differentiation is that both rely on Blue Origin's BE-4 engine.

Blue Origin has experienced significant delays in its development of the BE-4 engine. These delays have been cited as stemming from multiple causes, including insufficient engine hardware for testing, COVID-19 impacts, design changes, the need to meet stringent military requirements, and a lack of spare parts. Turbomachinery redesigns in 2019–2020 required reworking and retesting the engines. The military's certification process for combustion instability and other performance metrics is also cited as a source of delay.¹⁵ We have no independent knowledge that would allow us to confirm or repudiate these claims. We note that these delays occurred despite reported usage of advanced simulation tools meant to provide early risk reduction, indicating that rocket science remains challenging. Recently, new leadership has been brought in at Blue Origin.¹⁶

It is our assessment that New Glenn development is significantly affected by the NSS Phase 2 decisions. Vulcan's selection for NSS launch has likely incentivized Blue Origin to prioritize BE-4 development—a necessary and highly visible milestone—above that of the fully integrated New Glenn. However, we note that Blue Origin continues to invest in New Glenn development and the company has significantly matured its manufacturing capability in the past four years. Capital improvements at the facilities at Cape Canaveral; Huntsville, Alabama; and Seattle support the design, manufacturing, and launch of New Glenn. Additionally, Blue Origin has built a test facility in western Texas and has leased test Stand 4670 at the National Aeronautics and Space Administration's (NASA's) Marshall Space Flight Center to help in the testing of the BE-4 engine.¹⁷

Rocket Lab

Rocket Lab is a firm based in New Zealand and the United States. The company is one of the few space companies traded in the stock market and has a market capitalization of about \$2.1 billion as of November 2023. Rocket Lab is seeking to leverage its significant market share and experience in the small lift market to enter the heavy lift market. Its Electron launch vehicle serves the small lift market and has been used to execute 37 successful launches since 2017, including one in 2020 and two in 2022 for the National Reconnaissance Office.¹⁸ These two payloads were launched from New Zealand and mark the first use of non-U.S.-based launch sites for national security missions. The Electron was one of the first boosters to incorporate 3D-printed engines.¹⁹ Rocket Labs has experimented with recovering the Electron's first-stage booster either by helicopter or ocean splashdown.²⁰

¹⁵ Eric Berger, "Blue Origin's Powerful BE-4 Engine Is More Than Four Years Late—Here's Why," *Ars Technica*, August 5, 2021; Mike Wall, "Blue Origin's BE-4 Rocket Engine Exploded During June 30 Test: Report," *Space.com*, July 11, 2023a.

¹⁶ Eric Berger, "Jeff Bezos Finally Got Rid of Bob Smith at Blue Origin," *Ars Technica*, September 25, 2023b.

¹⁷ NASA, "Blue Origin Agreement Signals Rocketing Growth of Commercial Space," press release, April 17, 2019.

¹⁸ National Reconnaissance Office, "NROL-199," webpage, undated.

¹⁹ Umair Iftikhar, "Rocket Lab Celebrates 100th 3D Print of Its Rutherford Engine," *3D Printing Industry*, July 10, 2019.

²⁰ Mike Wall, "Rocket Lab Recovers Electron Booster from Pacific Ocean After Satellite Launch (Photos)," *Space.com*, July 19, 2023b.

Rocket Lab intends to expand into heavier payload classes with the development of the Neutron, which is being designed to launch payloads of 13,000 kg (28,660 lbs) to LEO or 1,500 kg (3307 lbs) to Mars and Venus.²¹ As such, the Neutron is designed to fill specific niches in the heavy lift launch market defined in Table 1.1. In 2021, Rocket Lab announced the receipt of a \$24 million award from the USSF to be used in maturing the Neutron engine design and declared its commitment to “becoming a launch provider for the National Security Space Launch (NSSL) program.”²² Rocket Lab’s prior collaboration with the National Reconnaissance Office demonstrated launch reliability and launch responsiveness and may allow the company to be competitive for NSS Lane 1 launches.

Relativity Space

Another launch startup, Relativity Space, aims to revolutionize traditional launch vehicle manufacturing through the extensive use of 3D printing and autonomous technologies. Its first launch vehicle, Terran 1, used 3D printing in the manufacture of a wide variety of components, as well as for the engines. This effort pushed the state of what is possible and used innovative metals developed at NASA.²³ Terran 1 completed a test flight in March 2023, with mixed success, before the company shifted focus to applying lessons from Terran 1 to its new launch vehicle, the Terran R.

Terran R is a two-stage launch vehicle being designed to deliver over 23,500 kg to LEO or 5,500 kg to geosynchronous transfer orbit in its reusable configuration, or 33,500 kg to LEO in its expendable one.²⁴ This design goal places it in direct competition with ULA’s Vulcan (27,000 kg to LEO) and the EU’s Ariane 6 (20,000 kg to LEO).²⁵ While the capability of the launch vehicle would support the range of NSS-required orbits, Relativity Space’s chief executive has explicitly stated that the lack of competition for SpaceX in the addressable market motivated its decision to focus on heavy lift.²⁶ This is not to say that Relativity Space does not intend to pursue NSS launch orders, only that the NSS market does not appear to be its primary objective. The projected first launch date is 2026. As of April 2023, Relativity Space valued the backlog of orders for its Terran R at \$1.65 billion.²⁷ Buyers placing multilaunch orders include Intelsat and OneWeb, illustrating that buyers are actively supporting new entrants in a bid to increase competition in the addressable market.²⁸

²¹ Rocket Lab, “Neutron: Mega Constellation Launcher,” webpage, undated.

²² Rocket Lab, “Rocket Lab Wins \$24M U.S. Space Force Contract to Develop Neutron Upper Stage,” press release, September 2021.

²³ NASA, “3D Printed Rocket Launched Using Innovative NASA Alloy,” May 2, 2023.

²⁴ Relativity Space, “Relativity Space Shares Updated Go-to-Market Approach for Terran R, Taking Aim at Medium to Heavy Payload Category with Next-Generation Rocket,” press release, April 12, 2023.

²⁵ ULA, “Vulcan,” webpage, undated; Arianespace, “Ariane 6,” webpage, undated.

²⁶ Jeff Foust, “Intelsat Signs Multi-Launch Contract with Relativity Space,” SpaceNews, October 11, 2023c.

²⁷ Relativity Space, 2023.

²⁸ Eutelsat OneWeb, “Relativity Space and OneWeb Sign Multi-Launch Agreement for Terran R,” press release, June 30, 2022.

Northrop Grumman/Firefly Aerospace Partnership

Northrop Grumman competed for the NSS Phase 2 procurement, proposing development of the OmegA launch vehicle, but was not selected. Development of the OmegA was subsequently dropped, and we assess that the loss of the NSS Phase 2 contract was a major factor in that decision, given that Northrop Grumman historically has competed only in U.S. government markets.

Northrop Grumman participates in the small lift launch market using Minotaur and Pegasus launch vehicles in support of U.S. government missions launching from a variety of launch complexes, including one at the Kwajalein Atoll in the Pacific Ocean, NASA's Wallops Flight Facility in Virginia, Pacific Spaceport Complex—Alaska on Kodiak Island, and Vandenberg Space Force Base in California. It also uses the Antares launch vehicle to serve the medium lift launch market, with a contract to fly cargo resupply missions using its Cygnus spacecraft to the ISS from its Wallops Flight Facility.²⁹ The Antares does not compete in the addressable market.

The Antares 230 first stage is designed and produced in Ukraine and powered by RD-181 engines from Russia. Production has been affected by supply chain disruptions in the aftermath of Russia's invasion of Ukraine in February 2022. As a result, Northrop Grumman has partnered with Firefly Aerospace to build a first stage for its next Antares model, the Antares 330.³⁰ To cover the gap between the launch of the last available Antares 230 (in August 2023) and the new Antares 330 (predicted for summer 2025), Northrop Grumman has contracted with SpaceX for three launches of the Cygnus payload to the ISS.³¹ It is unclear whether the Antares 330 could one day be a competitor in the NSS market. Currently, Northrop Grumman is focused on fulfilling its obligations under the cargo resupply contracts, and Firefly Aerospace sees the development of the first stage of the Antares 330 as a way to mature the development of its Miranda engine.

The Impact of NSS Acquisition Decisions on the NSS Market

As the 2019 assessment warned, "In many conceivable futures, the USAF's current acquisition plan is unlikely to provide a sufficient supply of NSS-certified launch vehicles to meet the projected demand."³² In fact, the USSF now finds itself in one of those futures. Shortages are all but inevitable for 2024–2026 given the backlog awarded to Vulcan under the Phase 2 contract. As shown in Figure 4.1, there is considerable supply risk for 2024–2026, independent of supplier selection or a decision by the USSF to support two versus three suppliers. While selecting three suppliers (to include the only currently certified supplier, SpaceX) would eventually bring sufficient manufacturing capacity online to reduce the uncertainty of risk of insufficient supply, it cannot eliminate the near-term impact of the Vulcan Phase 2 backlog (this case is shown in the right graph of Figure 4.1).

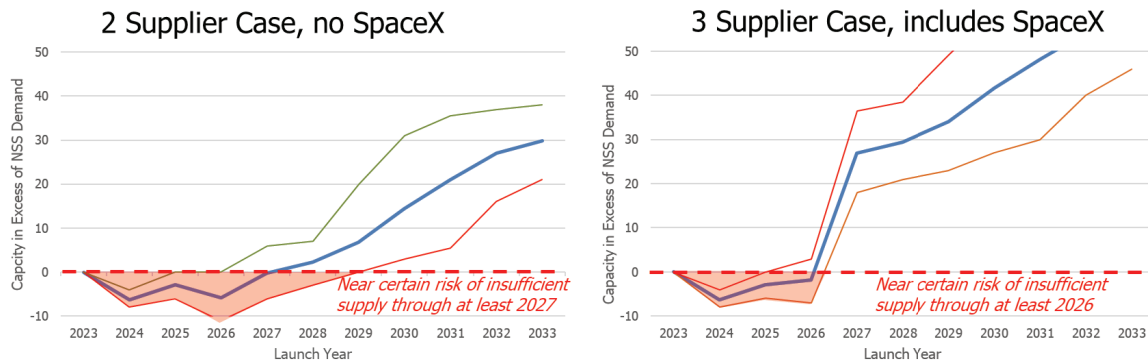
²⁹ Jeff Foust, "NASA Proposes Final Extension of ISS Cargo Contracts," SpaceNews, March 5, 2023a.

³⁰ Jeff Foust, "Northrop Grumman and Firefly to Partner on Upgraded Antares," SpaceNews, August 8, 2022b.

³¹ Josh Dinne, "Antares Rocket Makes Its Final Launch, Sending Cargo to the International Space Station," Space.com, August 1, 2023.

³² Triezenberg et al., 2020, p. 54.

Figure 4.1. NSS Supply Shortages Are Near Certain for 2024–2026



This undersupply is due to our assumptions regarding ULA’s build curve, which we graphically depict in Figure 4.2. Launch vehicles are complex and expensive to produce, integrate, and launch. When starting a new product line, firms must scale (i.e., build up) to full-rate production. A build curve describes the rate at which supply is produced during the buildup period. Consistent with the 2019 assessment, we assume that ULA’s build curve will equal that of the Falcon 9 in the early 2010s when it entered the market. To date, the Falcon 9 entry represents the fastest build curve in history for a launch service provider. There may be validity to arguments that ULA, unlike SpaceX in 2010 at the time of the Falcon 9 initial buildup, is an experienced launch provider that may be able to bring launches to market more quickly. Unfortunately, there is not yet evidence of ULA’s actual build rate, and we have no precedence for assuming a faster build curve. Furthermore, we note that the 24 launches per year by 2025 touted by ULA is beyond the company’s experience thus far.³³ At its peak in 2009, ULA provided 16 launches; in the most recent five-year period, annual launches have numbered ten or fewer.³⁴

Our build curve assumes that ULA’s production is not constrained by the availability of engines. This may be optimistic. Aerojet manufactures the engines for the Centaur upper-stage and recently stated that it needs to scale from its current 16–19 engines per year to 40 engines per year.³⁵ Blue Origin plans to scale production of the BE-4 to support both the New Glenn and Vulcan build curves.³⁶

Using the build curve uncertainties of Figure 4.2 to forecast Vulcan supply, we then project forward how long it is likely to take to burn down the NSS Phase 2 backlog. To simplify the calculations, we assume that all Vulcan production is dedicated to the already awarded NSS Phase 2 missions. Given ULA’s current orders from the addressable market, this is not a realistic assumption.

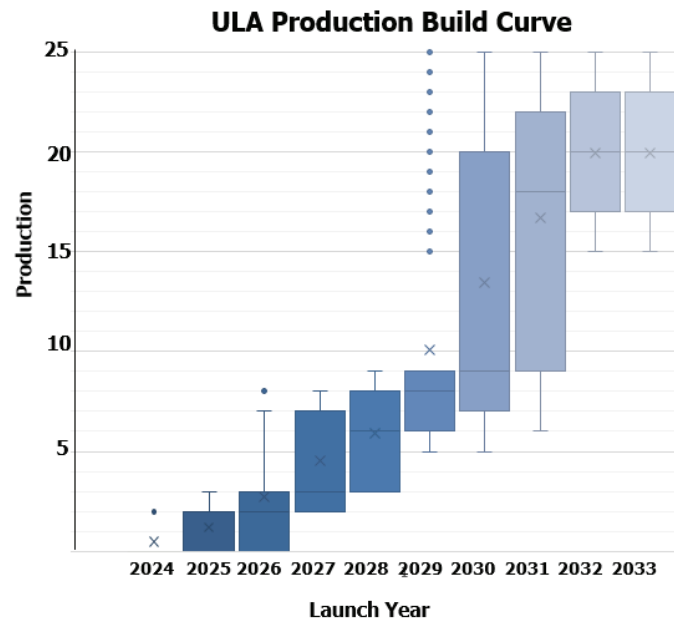
³³ ULA has stated this production goal multiple times—most recently in Andrew Jones, “See ULA Assemble New Vulcan Centaur Rocket for Its 1st Launch on Dec. 24 (Photos),” Space.com, November 27, 2023b.

³⁴ Triezenberg et al., 2020, Figure C.2, p. 101, and analysis of RAND’s Heavy Lift Launch Data Set, 2007–2023.

³⁵ Aerojet’s RL10 is also used on NASA’s Space Launch System, and Artemis launch vehicles and Vulcan will have to compete for priority with those more governmental uses. See Jeff Foust, “New RL10 Engine to Be Introduced on Vulcan in 2025,” SpaceNews, November 28, 2023e.

³⁶ We were unable to find recent public statements (i.e., those later than 2020) by Blue Origin regarding BE-4 production forecasts. As Blue Origin’s website currently states, “Full-rate production occurs in our engine manufacturing facility,” a future-looking statement with very little information; Blue Origin, “BE-4,” webpage, undated.

Figure 4.2. The Assumed Vulcan Build Curve



NOTE: This “box and whiskers” chart shows our predicted distribution of possible future production in each year. The solid bar (the box) depicts the values of the first and third quartiles, with a solid horizontal bar marking the median value. The lines extending from the solid bar (the whiskers) denote the values of the 5th and 95th percentiles, with outliers shown with an x.

Assumptions:

1. The first Vulcan launch occurs in late 2023 + [1 to 4 years]; NSS certification occurs in 2024 + [0 to 3 years].
2. The Vulcan factory is sized to support 15–25 launches per year.
3. Vulcan production is not constrained by BE-4 and RL10 engine availability.

The assumption for the sizing of the Vulcan factory is from discussions with ULA and its public statements. All other assumptions are the RAND team’s. The assumption on factory sizing is a driving factor only in years 2029 and onward. Prior to 2029, our assumptions on the uncertainty of a first launch date is the driving factor; after 2029, the uncertainty in the supplier’s choice of production level (within the bounds of the factory sizing) is the driving factor. The shading of the bars is simply to make the years more visually distinct and has no other meaning.

Therefore, this is a bounding analysis only. The results, shown in Figure 4.3, demonstrate that even under this bounding analysis, the last payload awarded to Vulcan under Phase 2 is likely to be launched in 2029 (versus the planned launch of 2026), and with 90-percent confidence, the last launch will be no sooner than 2027 and possibly as late as 2031. Due to the need to burn down the Phase 2 Vulcan backlog, we can state with virtual certainty that, under the assumptions of our model, NSS supply shortages will affect operational readiness through 2026 and are likely through 2029. We can also state with virtual certainty that this backlog will be of concern to commercial buyers who seek greater schedule certainty.³⁷

To put the risk of NSS shortages due to Vulcan backlog burndown into perspective, we calculate the probability of planned mission years of service that would be lost due to these shortages over the next ten years. We plot our results using the same risk framework used for the 2019 assessment. Figure 4.4 plots the results for three possible scenarios regarding the share of the FY 2024 awards

³⁷ As discussed in the 2019 assessment, buyers are hesitant to place orders with a supplier who has a significant number of NSS launches on its manifest. This is due to concerns regarding the schedule priority afforded to national security missions.

made to Vulcan—the contractual 60-percent share, a 40-percent share, and a zero share. (At the time we performed the analysis, the FY 2024 awards had not yet been made.) As can be seen, even this drastic step of awarding zero launches in FY 2024 to the Vulcan would not have removed the risk entirely. The consequence of earlier awards under the Phase 2 contract make that impossible.

The Phase 3 contract ordering period will cover FY 2025 to FY 2029. Future backlogs due to immature launch vehicles will be avoided under terms and conditions that allow the USSF to withhold Phase 3 awards to vehicles that are not NSS certified at the time of ordering.³⁸

The Impact of NSS Acquisition Decisions on the Global Addressable Market

To understand the impact of NSS acquisition decisions on U.S. providers that compete in the global heavy lift launch market, we expand our analysis with the assumption that Arianespace and one other foreign launch provider successfully bring a new launch vehicle to market. Given SpaceX's current monopoly position, there is a significant incentive to bring new vehicles into the global market; and China, in particular, has several existing vehicles with proven reliability. Over the past two years, China has matured both the Long March 5 and Long March 7, but has devoted all Long March 3, 5, and 7 production to its domestic market. Japan does not actively compete in the addressable market but is accepting launch contracts. Unfortunately, the initial launch of the H-III was not a success. India intends to compete, has proven launch vehicles, and has executed a handful of addressable launches. However, based on our analysis of the market, we assess that India has not yet crossed the chasm.³⁹ Russia, with historically proven expertise in launch, continues to supply those customers in the addressable global market who do not recognize sanctions imposed in response to its invasion of Ukraine.

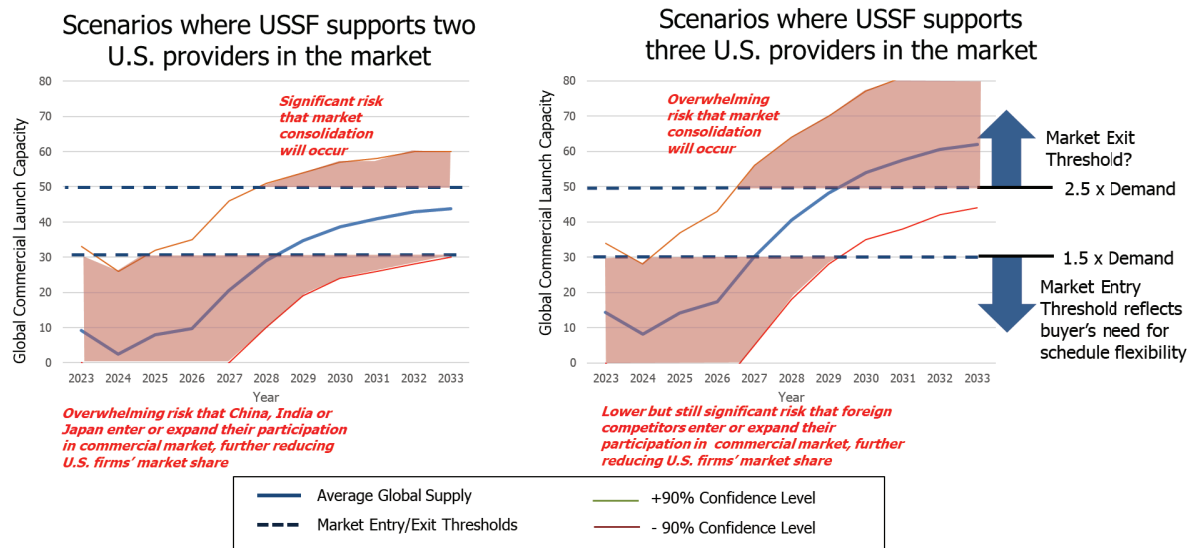
Our analysis is identical to that performed for the 2019 assessment, but with updated assumptions regarding the year that new vehicles will be brought to market. For U.S. firms, we include the planned capacity from SpaceX's Falcon family of launch vehicles, along with planned capacity from the following new entrants: Antares 330, Neutron, New Glenn, Terran R, and Vulcan. For each of the U.S. and foreign new entrants, we model their date of entry based on their most recent public statements of first launch date with a one- to four-year uncertainty. We also model a build curve for each (similar to that shown in Figure 4.2 for Vulcan) where the upper bound is based on each firm's forward-looking statements regarding factory sizing.

For each two-provider and three-provider case in which the USSF supports new entrants to enter the heavy lift launch market, Figure 4.5 plots average capacity that is available to meet addressable demand (i.e., in excess of NSS demand in the case of the new U.S. entrants, and in excess of both

³⁸ DAF, 2023a; DAF, 2023b.

³⁹ This statement is based on our observation that OneWeb, despite its compelling need for launch services following the invasion of Ukraine, only sparingly used the launch services of India's GSLV. Were the GSLV to be fully competitive in the market, especially given OneWeb's strategy of testing its business case by prioritizing the India-to-United Kingdom communications markets, we would have expected OneWeb to have used the GSLV more extensively.

Figure 4.5. Global Addressable Launch Capacity in Excess of Demand, 2023–2033



NOTE: This analysis assumes that SpaceX's Falcon 9 and/or Falcon Heavy remain in the addressable launch market independent of U.S. government support, but that unsupported new entrants do not. Capacity figures include Ariane 6 and a second foreign firm as it enters the market.

self-provisioned and NSS demand in the case of SpaceX). This capacity is then compared with two thresholds representing our assumed market entry and exit conditions.⁴⁰ As in the 2019 assessment, we set the market entry threshold at one and one-half times the average historic addressable demand, and the market exit threshold at two and one-half times the addressable demand. When capacity is below the market entry threshold, firms and nations are incentivized to enter the addressable market. When capacity is above the market exit threshold, consolidation is likely—too much supply will be chasing too little demand.

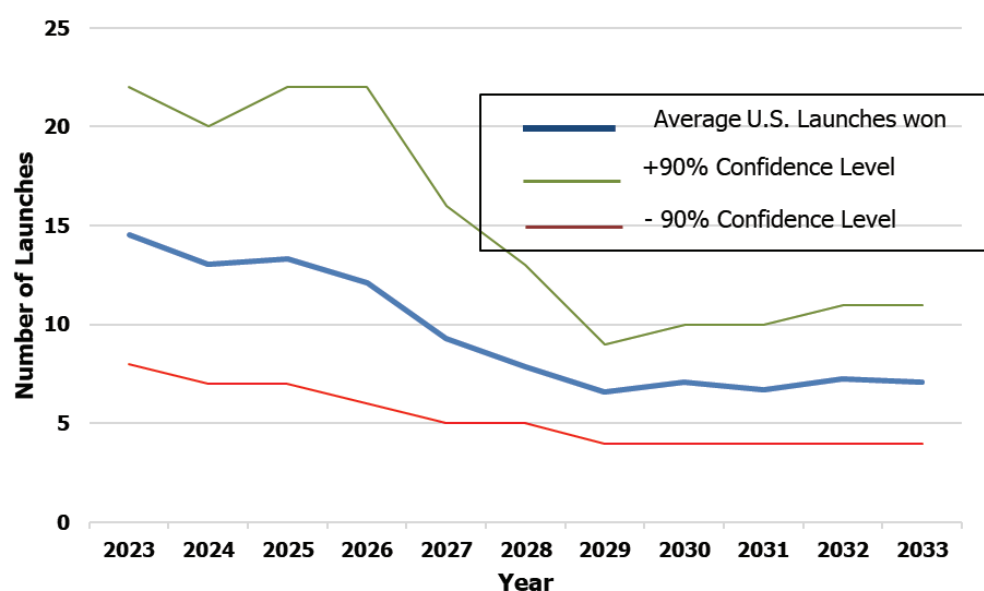
Both the risk of new entrants and the risk of market consolidation are an order of magnitude greater than that predicted in the 2019 assessment. Due to the current lack of competitors and the backlog of NSS Phase 2 Vulcan launches, there is a slight probability that supply to meet the addressable market demand will become negative in the next few years. Although this reality is inducing multiple U.S. firms to enter the market, we see no current indicators that firms in China, India, or Japan plan to more aggressively engage as sellers in the market. We believe that market consolidation will occur rapidly after one or two U.S. entrants have a successful first launch. Market consolidation may occur even earlier if news reports in early 2024 regarding a sale of ULA are correct.

The next step in our analysis is to determine the number of addressable launches likely to be awarded to U.S. providers as new foreign vehicles become available. Prior to a successful first launch of a foreign new entrant, we model the share of the addressable market available to U.S. firms as a

⁴⁰ Support for new entrants can come in many forms and should be tailored to each supplier's chosen market strategy. As is noted in Appendix C: Learning from History, in Triezenberg et al., 2020, pp. 97–114, government support appears to be essential not just for entry into the market but also for long-term survival. A discussion of conditions that led to previous consolidations in the launch industry can also be found in that appendix.

random variable between 30 and 90 percent of the addressable demand.⁴¹ After one new foreign entrant, we model U.S. market share as dropping to between 30 and 50 percent of demand. After two new foreign entrants, we model U.S. market share as falling to between 20 and 30 percent of demand. The results are shown in Figure 4.6. Through 2026, the number of launches expected to be awarded to U.S. firms has a very high variance—our 90-percent confidence levels are as high as 22 and as low as 6. After a foreign entrant succeeds, the uncertainty on our prediction shrinks. In the years 2029–2033, when it is more likely that two foreign entrants have succeeded in entering the market, the number of launches awarded to U.S. entrants is expected to be between four and 11 launches per year. Combined with the expected NSS demand of ten to 13 launches per year and a handful of U.S. national affinity launches, the total number of launches to be split among all U.S. providers would be in the range of 20 to 25. This quantity is unlikely to sustain more than two U.S. providers. The exact number of providers is dependent on each firm’s chosen strategy as it seeks to position itself in the market.⁴² In the long term, expectations that the USSF can leverage a thriving commercial launch market to meet its obligation to sustain two providers may not be realistic.

Figure 4.6. Projected Launches Won by U.S. Firms in the Global Addressable Market, 2023–2033



⁴¹ For the 2019 assessment, the assumed upper bound on U.S. market share was set at 67 percent because it was assumed that buyers in the market would act to prevent the United States from gaining a monopoly position. In fact, that did occur, and U.S. market share fell in the following years prior to the supply shock that came with the invasion of Ukraine. While buyers are clearly searching for viable alternatives to SpaceX’s monopoly, the reality is that until a new foreign vehicle enters the market, U.S. firms in the near term enjoy almost unfettered access to the addressable market demand.

⁴² For example, a firm with a large quantity of profit-generating, self-provisioned launches might be able to survive with only a very few launches from the NSS, national affinity, and global addressable markets. For a discussion of the various strategies firms have historically adopted when positioning themselves within the heavy lift launch market, see Triezenberg et al., 2020, Chapter 4, pp. 43–48.

Launch Resiliency Considerations

For this 2023 update, we were also asked to take a preliminary look at resiliency concerns related to U.S. launch sites. *Resiliency* is defined as the ability to continue to provide services for NSS launch when needed by the nation in the event of natural catastrophe or adversary attack. The RAND team had last examined this topic in an FY 2019 study, and our first step was to review that prior work to determine if there were any significant events in the intervening years, such as a loss of existing capacity or an unexpected increase in demand for launch site services, that would change those findings and recommendations.⁴³ We did not find any.

We then widened our scope to consider not just the resilience of the launch sites but that of the critical resources needed for launch. Table 4.1 provides information on key resources needed by current and new entrant U.S. launch vehicles. As demonstrated in the table, diversity of suppliers does not always lead to diversification of the key resources needed for launch. While the mutual dependence of both New Glenn and Vulcan on the BE-4 engine is widely known, other common supply chain dependencies are more difficult to ascertain due to a lack of public data regarding the supply chains for these launch vehicles. One item we sought to understand in our discussions with providers was whether there is a similar mutual reliance on 3D printing technology or supplies. While many of the newer launch vehicles do rely, or plan to rely, on 3D printing of engines, we observe that no de facto standard has emerged for the tools or materials needed in the 3D printing of engines. Therefore, it is our assessment that there is sufficient diversity in the practice of 3D printing that common-cause failures or supply shortages are unlikely to affect NSS launch in the next five years. As this is a quickly evolving area of manufacture, this assessment may not be valid a year from now.

Table 4.1. Key Resources Needed for Launch

Launch Service Provider	Key Suppliers	Fuel Needed	Launch Sites
Falcon 9/Falcon Heavy	SpaceX	Liquid oxygen (LOX)/ rocket propellant 1 (RP-1)	Kennedy Space Center, Cape Canaveral, Vandenberg
Vulcan	Blue Origin, Aerojet, Northrop Grumman	LOX/ liquefied natural gas (LNG)/ liquid hydrogen (LH ₂)/ solid propellant	Cape Canaveral, Vandenberg
New Glenn	Blue Origin	LOX/LNG/LH ₂	Cape Canaveral
Antares 330	Firefly, Northrop Grumman	LOX/RP-1/ solid propellant	Wallops
Terran R	Relativity Space	LOX/methane	Cape Canaveral
Neutron	Rocket Lab	LOX/methane	Wallops

SOURCE: RAND analysis of information available on manufacturers' websites.

⁴³ Gary McLeod, Ellen M. Pint, Eric V. Larson, Éder M. Sousa, and Jonathan Tran, *Assessing U.S. Space Launch Locations to Support the National Security Space Launch Program: An Independent Analysis of the Ability of the Eastern and Western Ranges to Support Forecasted Launch Demands*, RAND, 2019, not available to the general public.

We also looked at the types of fuel needed for each launch vehicle. All current and potential NSS providers are susceptible to a shortage of LOX, which occurred during the pandemic when COVID-19 hospitalizations, especially in Florida, diverted LOX from launch sites to hospitals to help patients on ventilators, thus affecting launches for NASA and ULA.⁴⁴ Nonetheless, we assess that such a shortage would be highly unlikely under more usual conditions. LOX is readily available in the United States, and the country is the top producer of LOX at almost 11 million metric tons per year at more than 110 facilities spread around the country, of which 1.2 million metric tons are exported.⁴⁵

Liquid fuels commonly used in launch vehicles include RP-1, methane or LNG, and LH₂. RP-1 is a type of kerosene that is highly refined for use in rocket engines, and its production requires specialized refineries to achieve the necessary level of purity. However, there are currently no anticipated shortages for these fuels in relation to the current vehicles. Moreover, the newest launch vehicles will transition to methane or LNG (which is between 85 and 95 percent methane and 5–15 percent ethane, propane, butane, and nitrogen).⁴⁶ Liquid methane offers easier handling and storage than RP-1, and nontoxicity, and can be used for a wide range of rocket applications.⁴⁷ The LNG supply chain is well established internationally, and the United States produces all the gas that it consumes, exporting the remainder. U.S. production capacity has been steadily growing over the last decade and is projected to continue growing in the near and long terms.⁴⁸ However, there are limited data available characterizing the methane supply infrastructure needed to support increased launch demand, and we recommend that the USSF assess the resilience of that infrastructure as it matures.

LH₂ utilization as a fuel in NSS launch vehicles is exclusively limited to the upper stages. While the utilization of LH₂ for rocket applications seems to be decreasing, its utilization is increasing for other applications—particularly in fuel cell vehicles.⁴⁹ In the near term we do not anticipate LH₂ supply shortages that could disrupt the provider's planned launch cadence.

Antares 330 and Vulcan are slated to use solid rocket motors (SRMs) as boosters. Both vehicles' SRMs will be supplied by Northrop Grumman. Decreased demand for SRMs, particularly after the retirement of the space shuttle program, resulted in excess capacity and consolidation of the supply chain. Over the last 20 years, the number of SRM manufacturers fell from six to two.⁵⁰ A 2017 Government Accountability Office report noted that limited new missile development programs

⁴⁴ Justin Bachman, "SpaceX, ULA See Launch Risk from Low Oxygen Supply Amid Pandemic," Bloomberg, August 28, 2021.

⁴⁵ U.S. Environmental Protection Agency, Oxygen Supply Chain—Executive Summary, undated.

⁴⁶ U.S. Department of Energy, *Liquified Natural Gas: Understanding the Basic Facts*, EO/FE-0489, August 2005.

⁴⁷ Todd Neill, Donald Judd, Eric Veith, and Donald Rousar, "Practical Uses of Liquid Methane in Rocket Engine Applications," *Acta Astronautica*, Vol. 65, Nos. 5–6, October–November 2009.

⁴⁸ U.S. Energy Information Administration, "Natural Gas Explained: Where Our Natural Gas Comes From," December 1, 2023.

⁴⁹ Ram R. Ratnakara, Nikunj Gupta, Kun Zhanga, Casimir van Doornea, James Fesmireb, Birol Dindorukc, and Vemuri Balakotaiah, "Hydrogen Supply Chain and Challenges in Large Scale LH₂ Storage and Transportation," *International Journal of Hydrogen Energy*, Vol. 46, No. 47, July 2021.

⁵⁰ Aerojet and Northrop Grumman (the latter through its acquisition of Orbital/ATK) are the two major producers of SRMs for defense applications.

reduce opportunities for SRM manufacturers to maintain critical design engineering skills.⁵¹ However, DoD's increased interest in hypersonic missiles and the heavy use of SRMs in the Russo-Ukrainian War has accelerated investments in SRM manufacturing that could spill over into launch vehicle propulsion systems.⁵² As such, we recommend that the USSF pay particular attention to changes in the SRM supply chain and how those changes might affect NSS launch providers, as this area remains dynamic. In just the past year, a new entrant, X-Bow Systems, has entered the SRM market, promising to reduce the cost of SRMs. DoD has taken notice and awarded contracts to the company.

We have one last note as we conclude this discussion of the fuels needed for launch services. The concentration of volatile fuels within a small geographic footprint and in a relatively well populated area, such as occurs at Cape Canaveral, has implications for range safety. There is current controversy over the calculation of equivalent tons of trinitrotoluene (TNT) for some of the new fuels and, depending on how the issue is resolved, the United States may be forced to reduce the concentration of launch complexes or stagger the amount of fuel that can be stored at the launch complexes to avoid creating an unacceptable hazard. In the future, the USSF may wish to consider diversity of launch sites or of fuel sources when awarding NSS launch contracts.

Summary

This chapter has provided in-depth descriptions of the changes on the heavy lift launch market over the past four years. In general, the NSS Phase 2 acquisition decisions had a significant impact on both the U.S. providers who were selected and those who were not. Furthermore, delays in bringing new vehicles to market and SpaceX's monopoly position have incentivized additional firms to seek to enter the heavy lift launch market. We assess that the structure of the NSS Phase 3 contract, with its allowance for unrestricted competition for Lane 1 launches, has been a factor in encouraging at least some of these new entrants. The most significant impact in the past four years, however, is the backlog of NSS launches on ULA's Vulcan manifest, as we will discuss in Chapter 5.

⁵¹ Government Accountability Office, *Solid Rocket Motors: DOD and Industry Are Addressing Challenges to Minimize Supply Concerns*, GAO-18-45, October 2017.

⁵² Assistant Secretary of Defense for Industrial Base Policy, "DoD Launches Effort to Enhance Solid Rocket Motor Production Capacity," press release, September 21, 2023; DoD, "DoD Strengthens Supply Chain for Solid Rocket Motors," press release, April 14, 2023.

Findings and Recommendations

In this chapter, we present our key findings and recommendations.

Key Findings

The average yearly addressable launch demand for heavy lift launch has remained relatively unchanged since the RAND team's 2019 assessment.¹ Demand rebounded in 2022 after a dismal two years in 2020 and 2021 and, while 2023 was better than average, the number of addressable launches made that year did not reach the postpandemic high seen in 2022. Demand is diversifying in terms of both the types of payloads being launched and their orbital destination. We expect to see slight overall growth in the next five years, driven by earth and space science payloads.

There are currently few viable competitors in the global addressable launch market; SpaceX's Falcon family of vehicles dominates, with a 70-percent share of the addressable demand in 2022 and a 92-percent share in 2023 (compared with a 40-percent share in 2019). Soyuz had been holding an equal or greater share to SpaceX's Falcon; that share was lost due to Russia's invasion of Ukraine in early 2022. International competitors that we thought might contend for market share have not; the reasons are varied. China's Long March family of vehicles appear to have been withdrawn from the international addressable market and now serve China's domestic demand only. Both the EU and Japan have retired their current launch vehicles but have either failed in their attempts to launch new vehicles (Japan's H-III) or delayed such an attempt (the EU's Ariane 6).

Given the current lack of competitors, several U.S. firms are seeking to enter the heavy lift launch market. However, in an era when demand is essentially flat, we expect rapid consolidation in suppliers should any become successful. While new entrants may help to diversify the overall U.S. industrial base of launch providers, we caution that there are shared common resources (fuel, as well as geographic concentration of launch complexes) that limit the impact of that diversity on overall mission resilience.

One of the largest impacts NSS decisions have had on the heavy lift launch market in recent years is the selection of SpaceX's Falcon family and ULA's Vulcan as NSS launch providers under the Phase 2 acquisition. That decision, and the backlog of 25 NSS missions that has built up while Vulcan works toward a successful completion of NSS certification activities, is likely to constrain Vulcan supply in the addressable market for several years. The last NSS payloads awarded under the Phase 2 contract are likely to experience a two-and-one-half-year delay even if all Vulcan production is

¹ In fact, average addressable demand fell slightly, from 20 to 19 annual launches.

dedicated to NSS.² If ULA makes some Vulcan capacity available to service its addressable demand, NSS missions may be delayed even further.

Recommendations

As recommended in the 2019 launch market assessment, the USSF should make prudent preparations for a future with only two U.S. providers of NSS certified heavy lift launch, at least one of which may have little support from the addressable marketplace. It is unlikely that other nations will allow U.S. firms to dominate the addressable market, and we do not foresee sufficient growth in the addressable or in U.S. nonaddressable demand to support more than two U.S.-based launch providers in the long term.

Despite the above recognition that the markets will not support more than two U.S. firms over the long term, we recommend that the USSF continue to provide support to enable as many U.S. launch providers as possible to enter the markets. We assess that the NSS Phase 3 launch contract, which allows for unrestricted competition for NSS Lane 1 missions, does this. By removing restrictions on competition for these missions, the USSF provides an incentive for U.S. firms to enter the heavy lift launch market without unnecessarily distorting that market. Regarding the USSF's decision to select up to three providers for NSS Lane 2 missions, we note that our analysis is largely agnostic. Doing so will certainly lead to earlier market consolidation, but that consolidation would occur regardless. Doing so may also deter foreign competitors from entering the market, but we see no evidence that foreign competitors factor that into their decisions.

The USSF has recognized the risks of adding to backlogs of firms that have not yet been certified for NSS launch and has mitigated that risk in the terms and conditions of the Phase 3 proposed contracts. While we are confident this step will allow the USSF to avoid worsening the existing situation, as noted above, the realized risk of the Vulcan Phase 2 mission backlog will affect Vulcan's participation in the addressable market for several years.

Finally, given the concentration of volatile fuels at Cape Canaveral, in particular, the USSF may wish to consider adding a diversity of launch sites or of fuel sources as a secondary consideration in future launch acquisition source selections to promote mission resilience.

² While multiyear NSS launch delays routinely occur due to demand slips, delays due to supply have occurred less frequently—the last being after the loss of the space shuttle *Challenger* in 1986.

Methodology

Data Sources

The RAND-developed heavy lift dataset documented in the 2019 report is the basis for the historic launch data used in our assessment.¹ We updated the dataset for the years 2020–2023 for this new assessment. Given that the Federal Aviation Administration no longer issues an annual review of satellite launches, we relied entirely on open-source data for the update. As in 2019, information from the Union of Concerned Scientists’ satellite launch data and RussianSpaceWeb.com were used.² We cross-checked against the list of launches recorded at the Space Launch Schedule website to provide secondary verification.³ Where discrepancies exist between these sources, we turned to open-source reporting from payload, space service, and launch service providers to resolve those discrepancies. Our methodology in determining the categorization, purpose, class of orbit, and operating entity is identical to that used in the 2019 assessment.

Analysis Overview

We project futures of the supply and demand of NSS and global addressable launch markets using Monte Carlo simulations, which are designed to estimate the distribution of a variable by letting it mimic the behavior of drawing many random samples from a specified known distribution. The benefit of simulating a distribution of many values (instead of a single value) is that distributions can be summarized to understand their expected value (typically, the arithmetic mean), their uncertainty (approximated by their variance), or their range (which we capture by looking at percentiles).

Sometimes researchers are particularly concerned with capturing their uncertainty in a variable’s distribution. Because Monte Carlo simulations have no limitations on what distributions can be chosen, researchers can fix any distribution’s hyperparameters to maximally capture their uncertainty. For our analysis, we draw randomly from continuous uniform distributions for the variables we are uncertain of. Because the uniform distribution has a variance that is proportional to the range of the distribution, we capture our uncertainty by adjusting the range of the distribution.⁴ We start by simulating variables (e.g., percentage of NSS launches delayed, or supplier capacity) with unique distributions—thereby assuming they behave independently. We then use these simulated variables to

¹ Triezenberg et al., 2020.

² RussianSpaceWeb.com, homepage, undated.

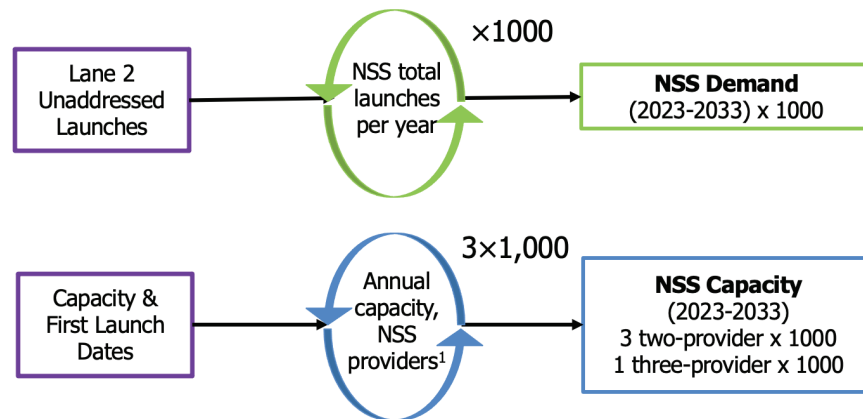
³ Space Launch Schedule, homepage, undated.

⁴ The formula is $Var(x) = \frac{(b-a)^2}{12}$ where $x \sim Uniform(a, b)$.

calculate the distributions of composite variables, such as total NSS launches per year or yearly supply.

To understand the NSS market, we first simulate baseline demand for NSS-certified launch services using the forecast shown in Figure 3.6. We then simulate baseline capacity for each possible NSS-certified launch provider. This process is summarized in Figure A.1 and provides the randomized data needed for subsequent analysis of the markets.

Figure A.1. Summarized Progression for Simulating NSS Demand and Capacity



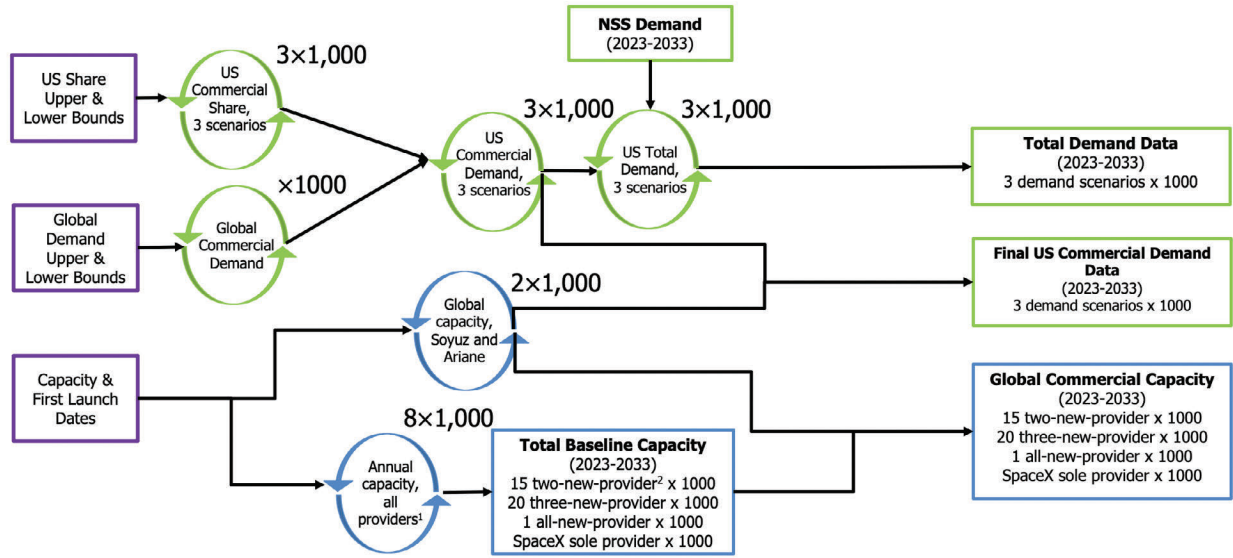
NOTE: ¹ NSS providers include those actively seeking, or who currently have, NSS certification: Blue Origin, SpaceX, and ULA.

We then expand the analysis to estimate the effect that the entry of new U.S. launch providers might have on consolidation of the global launch market. This requires projecting the capacities of new global entrants to the market and their effects on the total global capacities per year. We evaluate U.S. addressable demand under three scenarios depending on the behavior of the EU’s Ariane 6 and a second foreign entrant. U.S. addressable demand (in conjunction with the previous NSS demand scenarios) allows us to project annual U.S. total demand. The interactions between our assumptions and projected values are summarized in Figure A.2.

Throughout our simulations, we reduce the complexity of our results by reporting the arithmetic mean, the fifth percentile (e.g., –90 percent confidence), and the 95th percentile (e.g., +90 percent confidence).

Our assumptions on future demand are presented in Table A.1. As noted earlier in the report, demand is for the year of launch; for NSS demand, it is nominally two years after an award is made. Predictions of NSS demand for 2023–2026 are based on recent launch manifests and NSS Phase 2 awards; predictions for 2027–2031 are based on 100 percent of USSF projections for Lane 1 and 70 percent of USSF projections for Lane 2 in the Phase 3 Industry Day presentation; predictions for 2032–2033 project flat demand. Global addressable demand is drawn from a distribution, where the

Figure A.2. Summarized Progression for Simulating Addressable Demand and Global Capacity



NOTE: ¹ All providers are Northrop Grumman–Firefly’s Antares 330, Blue Origin’s New Glenn, ESA’s Ariane 6, Khronichev’s Angara (formerly Soyuz), Relativity Space’s Terran R, Rocket Lab’s Neutron, SpaceX’s Falcon, and ULA’s Vulcan.

² U.S. providers are Blue Origin, Northrop Grumman–Firefly, Relativity Space, Rocket Lab, SpaceX, and ULA.

Table A.1. Demand Projections, 2023–2033, Used in Market Analysis

Year	Unassigned NSS Launches	Global Addressable Demand, Lower Bound	Global Addressable Demand, Upper Bound
2023 ^a	0	21	27
2024	8	18	26
2025	4	19	29
2026	11	20	32
2027	7	18	32
2028	8	19	35
2029	9	17	35
2030	8	18	38
2031	8	16	38
2032	8	17	41
2033	8	15	41

^a The actual number of addressable launches for 2023 was 24.

lower and upper bounds are as shown in Table A.1 and graphically in Figure 3.5. These bounds are derived from a Monte Carlo simulation assuming a normal distribution of annual launches based on the cyclical behavior and linear growth in historical demand extrapolated into the future. The cyclical component is modeled as a cosine function, $A_{HIN} \cos \left[\frac{2\pi}{\tau_{HIN}}(t - t_0) \right]$, where the magnitude (A_{HIN}) and

frequency (τ_{HIN}) are set to mimic the historic demand set by the capital expenditure cycle of satellite communication service providers. As we note in the main body of the report, demand to support future proliferated satellite communication constellations might not exhibit this cyclical behavior, though there is some evidence that the refresh rates of providers like O3b and Iridium will be cyclical. The linear growth is modeled as $m_{MLIN} (t - t_0) - b_{MLIN}$, where the slope (m_{MLIN}) is set based on the recent increases in space and earth science missions. The intercept (b_{MLIN}) is based on launches in 2022.

We simulate launch capacity for each individual provider by assuming a date of first launch, a random delay, a capacity based on each provider's statements regarding how it is sizing its production resources, and a build curve. Each launch provider has unique assumptions for each of these values, as reported in Table A.2. In the table, the range from which we select a uniform distribution is shown in square brackets.

Table A.2. Capacity Assumptions for All Heavy Lift Launch Providers Modeled

Launch Vehicle	Projected First Launch Date	Projected First Launch Delay (δ)	Maximum Capacity (yearly launches)
Vulcan	2024 ^b	+ [0–3] years	[15–25]
New Glenn	2024	+ [1–4] years	[12–24]
Falcon^a	Not Applicable	None	[20–35]
Ariane 6	2024	+ [1–4] years	[5–11]
Second foreign launch vehicle	2025	+ [1–4] years	[8–14]
Neutron	2025	+ [1–4] years	[8–14]
Terran R	2026	+ [1–4] years	[8–14]
Antares 330	2024	+ [1–4] years	[8–14]

^a Falcon capacity is not inclusive of that needed for self-provisioned Starlink launches.

^b Actual Vulcan first launch was January 2024; a second launch is needed for NSS certification.

NSS Market Simulations

Simulation of NSS baseline supply and demand proceeds as it did in the 2019 analysis. For this update, we consider Blue Origin, SpaceX, and ULA as NSS providers when computing our NSS supply baseline. SpaceX's Falcon 9 and Falcon Heavy are certified for NSS launch, while Blue Origin's New Glenn and ULA's Vulcan are currently seeking NSS certification.

NSS Baseline Demand

To simulate NSS demand for one year, we generate a random number of delayed launches from a uniform distribution ranging from 0 to 30 percent of the year's unassigned launches (the first column of Table A.2). The year's unassigned launches are then adjusted to account for the number of delayed

launches from that year and the number of delayed launches from the previous year. For a single year, y , and simulation, i , this appears as follows:

$$\begin{aligned} \text{Adjusted Unassigned Launches (Year = } y)_i &= \\ &= \text{Unassigned Launches (Year = } y) - \text{delayed launches (Year = } y)_i \\ &+ \text{Delayed Launches (Year = } y - 1)_i \\ \text{where Delayed Launches(Year)} &\sim \text{Unif}(0, \text{Unassigned Launches (Year)} \times 30 \text{ percent}) \end{aligned}$$

We repeat this process 1,000 times for each year to generate a total of 11,000 simulations from 2023 to 2033. Because every year received 1,000 simulated values, we treat those as each year's hypothetical distribution. We summarize each year's distribution using its mean, 5th percentile, and 95th percentile.

Supply Under New Domestic Entrant Scenarios

We simulate launch capacity for each individual provider by assuming a date of first launch, a random delay, a capacity based on each provider's statements regarding how it is sizing its production resources, and a build curve. For one simulation of a provider's production for 2023–2033, we start by selecting the number of years a first launch will be delayed (δ) from a random uniform distribution. Each launch provider receives a unique distribution that is bounded by the assumed range listed in the third column of Table A.2.

We then select the provider's maximum capacity from a uniform distribution that is bounded by the assumed range in the fourth column of Table A.2. This range is based on firms' public statements regarding choices they have made or intend to make in positioning themselves in the launch markets. For example, ULA has stated that it is sizing its Vulcan factory to 25 annual launches. Therefore, we simulate that it will choose an annual production rate between 15 and 25 launches, inclusively, based on actual future demand and its overall market strategy.

However, only established launch providers are assumed to operate at this chosen capacity. For new entrants, we assume that capacity begins at the provider's first launch date and builds up to the chosen capacity—that is, we assume that new entrants will experience a buildup period where they operate at up to one-third of maximum capacity for their first three years and reach maximum capacity after five years. This assumption is based on the best case observed historic capacity build curve for a launch provider. Although it is possible that one of the new entrants could exceed this best-case scenario, we have no evidence on which to base a more optimistic projection. Launch buildup is also subject to uncertainty in the first launch date, which shifts the buildup period δ years from the currently projected first launch date (the second column in Table A.2).

Once capacity has been adjusted to account for the delay and buildup period from 2023 to 2033, we have concluded one round of simulation of the launch provider's capacity. To simulate NSS supply for 2023–2033, we generate capacities for the Falcon family of launch vehicles, New Glenn, and Vulcan by repeating the above process 1,000 times each.

With simulated capacities for each of the NSS launch providers, we can now observe the behavior of NSS supply under a variety of scenarios. Namely, we concern ourselves with how many NSS providers supply the launch market at the same time. Because there are three possible providers, there

are seven possible scenarios: three one-supplier cases (Blue Origin, SpaceX, or ULA); three two-suppliers cases (Blue Origin–SpaceX, ULA–Blue Origin, or ULA–SpaceX); and a single three-supplier case (ULA–Blue Origin–SpaceX).

Global Addressable Market Simulations

We now turn to our attention to the global launch market. We concern ourselves with the entrance of new U.S. launch providers and the expected U.S. share in the global market so we can estimate total and addressable U.S. demand.

U.S. Demand Under Three New Global Entrant Scenarios

Our process for estimating U.S. addressable demand begins by using annual estimated ranges of global demand (presented in Table A.1 and Figure 3.5) to generate 1,000 randomly uniform values of total global demand for each year from 2023 to 2033.

Simultaneously, we consider three scenarios of U.S. share in the global market. They account for high U.S. market share, where neither the Ariane 6 nor a second foreign launch supplier's vehicles enter the market; partial market share, where either foreign supplier enters the market; and low market share, where both enter the market. In each scenario, we assume a range of shares for the United States; these are presented in Table A.3. As with previous ranges, we treat each range as the bounds of a uniform distribution and generate 1,000 possible values for each scenario and each year.

Table A.3. U.S. Addressable Market Share Scenarios

#	Scenario	Projected U.S. Addressable Market Share
1	Prior to foreign entries	[30–90%]
2	After entry of one foreign launch vehicle	[30–50%]
3	After entry of a second foreign launch vehicle	[20–30%]

To calculate a possible value, i , of addressable demand won by U.S. firms for each year, y , dictated by each of the three scenarios, s , we start by using the following equation:

$$US\ Commercial\ Demand(Year = y)_{s,i} = US\ Commercial\ Share_{s,i} \times Global\ Commercial\ Demand(Year = y)_{s,i}.$$

With values from all three scenarios, we choose annual values of U.S. addressable demand depending on the annual capacity of the foreign launch vehicles. For a given year and simulation, if either have nonzero capacities, we use the simulated U.S. addressable demand values from Scenario 2 (the case where one foreign launch vehicle has entered the market). If both have nonzero capacities, we use the simulated U.S. addressable demand values from Scenario 3. In all other cases, we use U.S. addressable demand values from Scenario 1.

This brings us to the following equation for simulation i , year y , and scenario s :

$$\text{U.S. Addressable demand}(\text{Year} = y)_i = \begin{cases} \text{U.S. Addressable demand}(\text{Year} = y)_{\text{Low Market Scenario},i} & \text{if } \text{Ariane6 Capacity}(\text{Year} = y)_i \neq 0 \cap \text{Soyuz Capacity}(\text{Year} = y)_i \neq 0 \\ \text{U.S. Addressable demand}(\text{Year} = y)_{\text{Partial Market Scenario},i} & \text{if } \text{Ariane6 Capacity}(\text{Year} = y)_i \neq 0 \cup \text{Soyuz Capacity}(\text{Year} = y)_i \neq 0, \\ \text{U.S. Addressable demand}(\text{Year} = y)_{\text{High Market Scenario},i} & \text{if } \text{Ariane6 Capacity}(\text{Year} = y)_i = 0 \cap \text{Soyuz Capacity}(\text{Year} = y)_i = 0 \end{cases}$$

Finally, recall that we simulated 1,000 values of NSS addressed demand in the previous section. In conjunction with these NSS demand values, we can now use the U.S. addressable demand values to estimate U.S. total Demand for each of the three scenarios:

$$\text{US Total Demand}(\text{Year} = y)_{s,i} = \text{US Commercial Demand}(\text{Year} = y)_{s,i} + \text{NSS Demand}(\text{Year} = y)_i.$$

The results of the analysis are plotted in Figure 4.6 for the years 2023–2033.

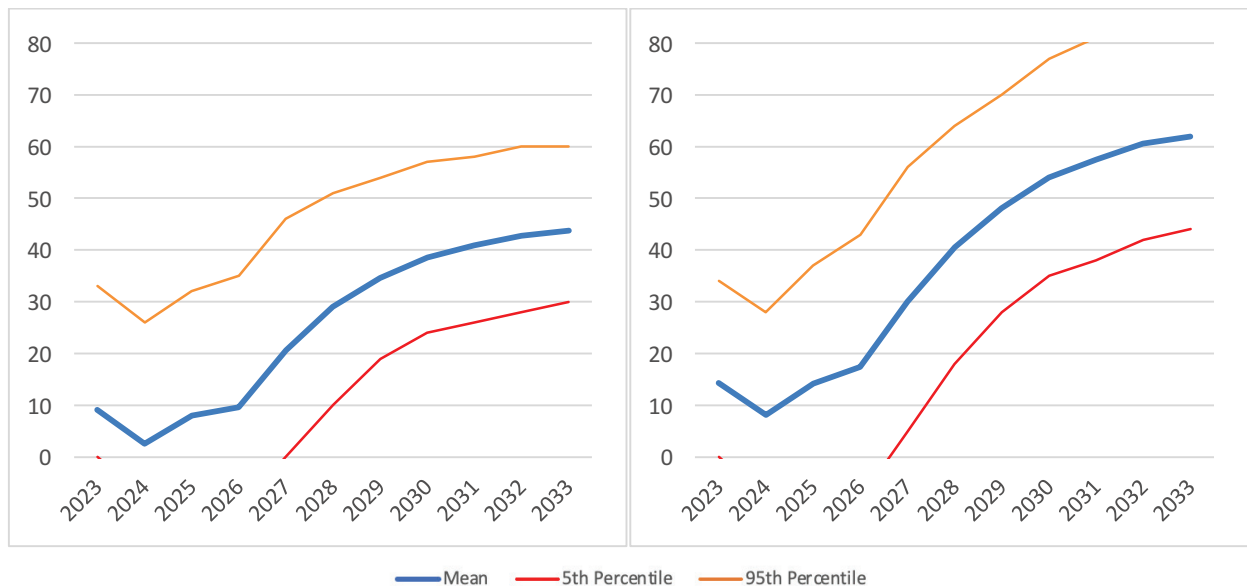
Global Capacity with New Entrants and the Implications for U.S. Addressable Demand

We simulate capacity for each of the global launch providers in the same fashion as the baseline NSS providers: using uniformly random delays, launch capacity bounds, and first launch dates.

For total global supply, we expand from NSS providers (Blue Origin, SpaceX, and ULA) to include all suppliers shown in Table A.2. We estimate capacity by using simulated values of delay, launch capacity, and first launch date. The supplier capacities are then added together in combinations for two types of markets: one where only two U.S. suppliers operate, and one where three U.S. suppliers operate. We also include one scenario with all six American suppliers. Finally, to arrive at global capacity, we add in the simulated capacities of the foreign providers.

Figure A.3 compares average launch capacity (the blue line) across all two-provider scenarios (on the left) and all three-provider scenarios (on the right). These are the results analyzed in Figure 4.1.

Figure A.3. Global Annual Capacity for Addressable Launch for All Two-NSS Provider Scenarios (Left) and All Three-NSS Provider Scenarios (Right)



Abbreviations

CI	confidence interval
COVID-19	coronavirus disease 2019
DAF	U.S. Department of the Air Force
DoD	U.S. Department of Defense
ESA	European Space Agency
EU	European Union
FFRDC	federally funded research and development center
FY	fiscal year
GEO	geosynchronous earth orbit
GPS	Global Positioning System
GSLV	Geosynchronous Satellite Launch Vehicle
IRL	innovation readiness level
ISS	International Space Station
lbm	pound-mass
LEO	low earth orbit
LH ₂	liquid hydrogen
LNG	liquified natural gas
LOX	liquid oxygen
MEO	medium earth orbit
NASA	National Aeronautics and Space Administration
nmi	nautical mile
NSS	national security space
PAF	Project Air Force
PSLV	Polar Satellite Launch Vehicle
RP-1	rocket propellant 1
SpaceX	Space Exploration Technologies Corporation
SRM	solid rocket motor
TRL	technology readiness level
ULA	United Launch Alliance
USAF	U.S. Air Force
USSF	U.S. Space Force

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The U.S. military depends on space-based assets for strategic and tactical communications, for collection of imagery over contested areas, for warning of missile launches, and for precision navigation and timing. The ability to replenish those assets in a timely fashion may affect overall force readiness. The U.S. Space Force (USSF) is required to sustain at least two viable U.S.-based launch service providers for national security space (NSS) payloads. In 2019, a RAND team assessed the global heavy lift launch market to understand the impact of NSS acquisition decisions on that market. Space Exploration Technologies Corporation (SpaceX) and United Launch Alliance (ULA) are the current NSS launch providers, having been selected for the Phase 2 launch contracts awarded between fiscal years 2020 and 2024. Although ULA is retiring its prior NSS-certified launch vehicles, its new vehicle has yet to complete NSS certification, leaving the USSF temporarily reliant on SpaceX as the only NSS-certified launch service provider.

The RAND team was asked to update its assessment of the launch markets to better understand how the USSF might shape future acquisition strategies or prepare for operational risks introduced by past acquisition decisions. The authors examine historical and projected levels of supply and demand in the global and NSS heavy lift launch markets. They forecast demand over the next ten years (2024–2033) and explore how assumptions regarding future market dynamics might affect the USSF's ability to meet NSS demand and sustain at least two certified suppliers.

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